CAPSTONE PROJECT 2 — MACHINE LEARNING

TENDERLENS: SINGLE-BID RISK IN ASSAM GOVERNMENT TENDERS

A Machine Learning Capstone Project Using Assam Government E-Tender Data, 2016–2023

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TOOLS USED

Python · pandas · SQL (SQLite) · scikit-learn · XGBoost · SHAP · Sentence Transformers
TensorFlow / Keras · Streamlit · Plotly · LLM API (Gemini / Groq)

Executive Summary

When a government tender receives only one bid, there is no competition. The department cannot compare prices, and the single bidder can quote close to the full estimate. In Assam's government e-tenders, this happened far more often than it should. Of the 6,790 tenders with a verified bidder list, 2,044 (30.1 percent) received exactly one bid. These single-bid tenders also carried more than their share of public money. Among tenders with a recorded value, they made up 31.3 percent of tenders but 36.5 percent of the estimated value, about Rs 14,752 crore. This project set out to predict which new tenders are likely to receive only one bid, at the moment they are published, so that reviewers know where to look first.

The analysis draws on 39,500 tender files from the Assam e-tenders portal, published by CivicDataLab and covering April 2016 to May 2023. After cleaning, these reduced to 39,411 unique tenders, of which 6,790 had a verified bidder list. The data was parsed and cleaned in Python, explored in SQL, and turned into seven publish-day clues, using only information available on the day a tender is published. To mirror real use, the data was split by time rather than at random:

- training on tenders up to September 2022
- validation on October to December 2022
- one final test on January to May 2023

A rule-based benchmark, Logistic Regression, Random Forest, XGBoost, two text methods (TF-IDF and sentence embeddings) and a neural network were compared against random checking. The final model is explained with SHAP. An Isolation Forest flags unusual tenders. The results are delivered through an eight-page Streamlit application that includes a GenAI review brief written only from verified facts.

The most important finding of this project is that tender patterns changed sharply after September 2022. Twenty-one of thirty-four clues reversed direction between the training and validation periods, so a model that learns from the past must be tested honestly against the future. On the one final test, carried out once with the plan locked in advance, the model's riskiest 20 percent of tenders contained 101 single-bid tenders, compared with 79 found by random checking. The model beat random checking in 95.8 percent of 500 re-draws: a small but real improvement. A simple rule based on re-tendering and department history found 108, and this strong baseline is reported openly rather than hidden. A neural network was also tested, but it only tied random checking (0.286 against 0.288 on validation), so the simpler and explainable Logistic Regression was kept. Complexity had to earn its place, and in this data it did not.

Alongside this, the data shows several further patterns:

- Real price discounts appear only when four or more firms bid. With four or more bids, the median award was 90 percent of the estimate; with two or three bids, it stayed close to the full estimate.
- Re-posting a tender did not attract more competition.
- Some departments moved against the overall trend. The Bodoland Territorial Council rose from 24.7 to 39.9 percent single-bid tenders after September 2022, which explains where the model missed.
- The anomaly checks also exposed a likely data-entry error: a solar power plant recorded at a value of Rs 3,000.

The report that follows documents the full pipeline behind these findings, from raw tender files to a tested, interactive application. It also records the checks used to prevent data leakage, the limitations of the data, and four practical recommendations. TenderLens is designed as screening support to help reviewers prioritise their work. It is not a fraud detection system.

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1. Introduction

Every year, Indian state governments spend thousands of crores of rupees through public tenders. A department that needs a road repaired, a building constructed, or medicines supplied publishes a tender, firms submit bids, and the work is normally awarded to the best qualified, lowest-priced bidder. The entire system rests on one simple idea: when several firms compete, the government gets a fair price.

That idea breaks down when only one firm bids. A single-bid tender has no competition at all. The department has nothing to compare the price against, and the one bidder has little reason to offer a discount. Public procurement research and international bodies such as the OECD treat single bidding as one of the most widely used warning signs of weak competition. It does not prove that anything went wrong, since a tender may simply be small, specialised, or poorly advertised, but it is a strong signal that a tender deserves a closer look.

This project studies single bidding in the Government of Assam's e-procurement system, using tender-level data published openly by CivicDataLab from the Assam e-tenders portal. The dataset covers tenders published between April 2016 and May 2023 across more than one hundred departments, and for the more recent tenders it records how many firms actually bid. This makes it possible to ask a practical question that most published analyses only describe after the fact: can single-bid tenders be predicted before the bids come in?

The output of this project is TenderLens, a machine learning pipeline and an eight-page interactive Streamlit application. The pipeline moves from raw tender files through cleaning in Python, exploratory analysis in SQL, feature engineering, model comparison, explanation with SHAP, anomaly detection with an Isolation Forest, and a deep learning comparison. The application presents the results as a ranked review list, a detailed explanation for every tender, and a short plain-English review brief written by a large language model. Every stage of that pipeline, and the reasoning behind it, is documented in this report.

2. Problem Statement

Which newly published Assam government tenders are likely to receive only one bid, why, and how can reviewers use that knowledge to decide which tenders to check first?

In the labelled Assam data, almost one in three tenders (30.1 percent) received a single bid, and these tenders together carried about Rs 14,752 crore of estimated value. Audit and vigilance teams cannot examine every tender in detail; in a busy month such as December 2022, more than 1,200 labelled tenders were published. Today, a single-bid outcome is usually discovered only after the bids are opened, when it is too late to re-advertise, extend the deadline, or invite more firms.

The problem this project addresses is therefore one of prioritisation. If a model can rank new tenders by their risk of attracting only one bidder, using only the information available on the day of publication, reviewers can spend their limited time where competition is most likely to fail. The model must also explain each ranking in simple terms, because a reviewer cannot act on a score they do not understand, and it must be tested honestly on future tenders rather than on a random sample of the past.

3. Objectives

The project was guided by nine core questions, each of which is answered in a later section of this report and, where relevant, on a page of the TenderLens application:

- How common are single-bid tenders in Assam, and how has this changed over the years?
- Which departments, types of work and times of year see the most single-bid tenders?
- How much public money flows through single-bid tenders, and do single-bid works receive smaller price discounts?
- Can single-bid tenders be predicted on the day they are published, better than random checking and a simple rule?
- Does the text of a tender (its title and description) add useful information beyond the basic tender details?
- Which clues push a tender's risk up or down, and can each prediction be explained in plain language?
- Which tenders look unusual compared with normal tenders, and do unusual tenders overlap with high-risk ones?
- Does a deep learning model earn its place over a simpler, explainable model?
- How can the results be delivered as a practical tool that a reviewer can use every month?

Beyond answering these questions individually, the project aimed to combine honest model evaluation, explainable results and a working application into a single, presentation-ready product: the kind of deliverable a real data science team would hand to an audit or vigilance department, not just an academic exercise.

4. Scope and Limitations

This project covers tenders published on the Government of Assam e-tenders portal between April 2016 and May 2023, as made available by CivicDataLab. It does not include other states, central government tenders, or tenders awarded outside the e-procurement system. Only one state was studied so that every finding could be checked carefully against the raw data.

Of the 39,411 unique tenders, only 6,790 carry a verified list of bidders. Older tender files simply do not record bids, so the prediction model is built and tested only on these 6,790 labelled tenders. Tenders with an unknown number of bids were never treated as multi-bid; they were kept as unknown. Most labelled tenders come from 2022 and early 2023, which means the model learns from a relatively short history.

The model is restricted to information available on the day a tender is published. Facts that only appear later, such as the number of bids, the winning firm, the award price, cancellations and amendments, were never used as model inputs. They were analysed separately to understand the bigger picture of competition and price.

Finally, TenderLens is a screening tool. A high risk score means that a tender looks similar to past tenders that received only one bid. It does not mean that anything improper happened, and the project deliberately avoids the word fraud. Supplier names are anonymised throughout the report and the application.

5. Tools and Technologies

The project follows a staged pipeline, with each tool used for the task it is best suited to rather than forcing all the work into a single environment. All analysis was carried out in Jupyter notebooks inside VS Code on a MacBook, with Python 3.11.

Stage	Tool	Purpose
Data audit and cleaning	Python, pandas, NumPy	Reading 39,500 tender CSV files, joining them into one table, fixing data types, dates, duplicates and odd values
Exploratory analysis	SQL (SQLite database), pandas	Grouping and comparing single-bid rates by year, department, value, bidding window and re-tendering
Visualisation	Matplotlib, Seaborn, a shared project style file	85 charts in one consistent colour scheme, each titled with its finding
Machine learning	scikit-learn, XGBoost	Logistic Regression, Random Forest, XGBoost, Isolation Forest, scaling and evaluation metrics
Text analysis (NLP)	TF-IDF (scikit-learn), Sentence Transformers (all-MiniLM-L6-v2)	Turning tender titles and descriptions into numbers and testing whether wording predicts single bids
Explainability	SHAP, permutation importance	Showing which clues push each tender's risk up or down
Deep learning	TensorFlow 2.16 with Keras 3	A neural network classifier and an autoencoder, compared fairly against the simpler models
Application	Streamlit, Plotly, fpdf2	The eight-page TenderLens app with interactive charts and downloadable PDF briefs
Generative AI	LLM API (Google Gemini or Groq), with a template fallback	Writing a short plain-English review brief for any tender, using only its verified facts
Version control	GitHub	Storing notebooks, app code, models and reports

In short: Python and pandas handled the heavy cleaning of a large government dataset, SQL supported the exploration, scikit-learn and TensorFlow powered the models, SHAP made them explainable, and Streamlit turned the results into a tool a reviewer can actually use.

6. Methodology

The work was organised into ten stages, each in its own notebook. Every stage built on the files saved by the one before it, so the whole pipeline can be re-run from the raw data. The table below summarises the flow.

Stage	Notebook	What it produced
S1	01_data_audit	Understanding of the raw file structure and the data funnel
S2	02_cleaning	One clean table of 39,411 tenders and 39 columns, with a single-bid label
S3	03_eda	Patterns by year, department, value, bidding window and re-tendering, with SQL queries
S4	04_feature-engineering_split	Seven publish-day clues and a time-based train, validation and test split
S5	05_models	Benchmarks and three model families on validation; discovery of drift
S6	06_nlp	Text methods and the label timing check
S7	07_final_model	The one final test, risk bands and SHAP explanations
S8	08_anomaly_patterns	Money at risk, competition and price, Isolation Forest, check-first rule
S9	09_deep_learning	Neural network and autoencoder comparison
S10	Streamlit app	The TenderLens application with the GenAI review brief

6.1 Data Collection

The data comes from CivicDataLab's public Assam tenders repository on GitHub, which collects tender pages from the Assam e-tenders portal. It was downloaded as a single 72 MB zip file containing 39,500 CSV files, where each file describes one tender. The official Open Contracting (OCDS) feed was also considered but not used, because it does not contain bid counts or award details.

The file structure needed careful study before any analysis. In each file, the first row holds the tender details (department, title, value, dates and fees), while any extra rows hold lists such as bidders and documents. The 30,291 older files contain no bidder list at all. The 9,209 newer files, whose names start with "final_", contain the bidder information. In awarded tenders, one column holds only the winner's name and a second column holds the full list of bidders, so the number of bids had to be counted from the second column.

6.2 Data Cleaning

All 39,500 files were joined into one master table and cleaned in sixteen documented steps. Each step was logged with the number of tenders remaining, so the cleaning can be traced from start to finish. The main issues found and the decisions taken were:

- **Duplicates:** 89 tender IDs appeared twice. The newer copy was kept, leaving 39,411 unique tenders.

- **Types and dates:** seven number columns stored as text were converted to numbers, and three date columns (published, last date and bid opening) were converted to real dates.
- **Bidding window:** 19 tenders had a negative number of days to bid and were set to empty; 129 very long windows were marked rather than deleted.
- **Tender value:** a value of Rs 0 means the value was not written, so it was set to empty and a separate *value_missing* flag was added. Six very large values above Rs 1,000 crore were kept, because most were real projects, but they were marked.
- **Award ratio:** the awarded price divided by the estimate was calculated; ratios below 0.2 or above 5 were treated as entry errors and set to empty.
- **Department names:** the portal often adds sub-office details after a "||" sign. Keeping only the main name reduced 454 variants to 105 departments.
- **Data gap:** a gap in the portal data between March 2021 and February 2022 (489 tenders) was marked so that it would not be mistaken for a real drop in tenders.
- **Re-tenders:** 6,092 tenders were marked as re-tenders, either because the title said so or because the same department had posted the same work before.
- **Label:** the target *single_bid* was created as 1 for exactly one bid and 0 for two or more. This gave 2,044 single-bid tenders and 4,746 multi-bid tenders; all other tenders stayed unknown.

6.3 Exploratory Data Analysis and SQL

The clean table was loaded into a SQLite database so that patterns could be checked with SQL queries as well as with pandas. The exploration looked at single-bid rates by year, department, tender value, length of the bidding window, and how soon a tender was re-posted. The single-bid rate fell steadily over the years, from about 41 percent in 2021 to 22 percent in the first months of 2023 (Figure 1), and it varied widely between departments, from 51 percent for Food and Civil Supplies to 15 percent for Health and Family Welfare (Figure 2).

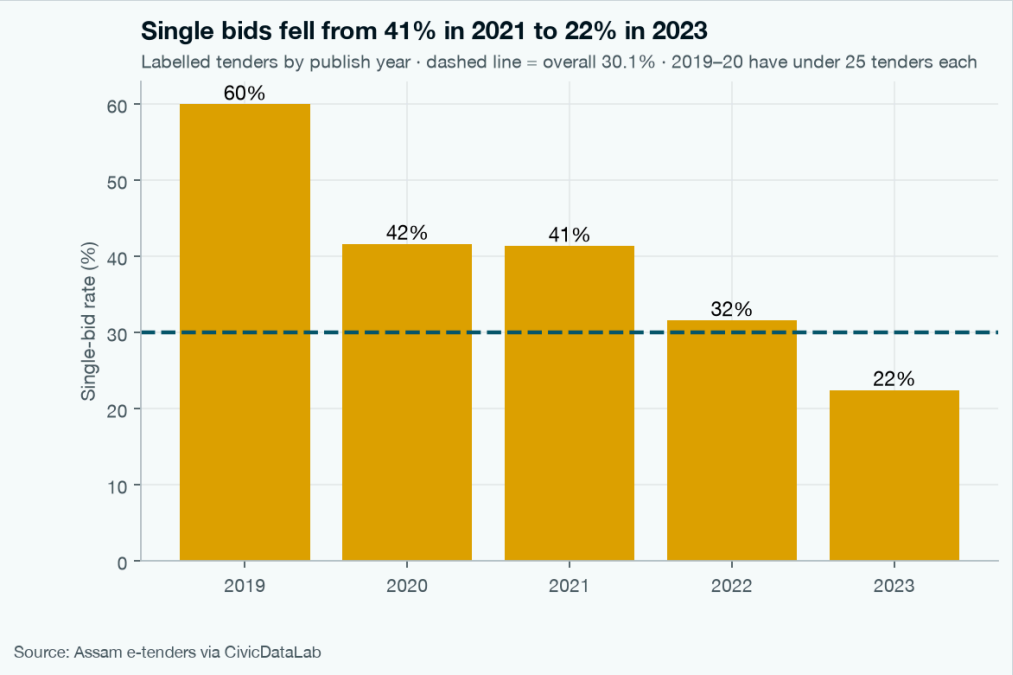


Figure 1 — Single-bid rate by year of publication (dashed line = overall 30.1 percent)

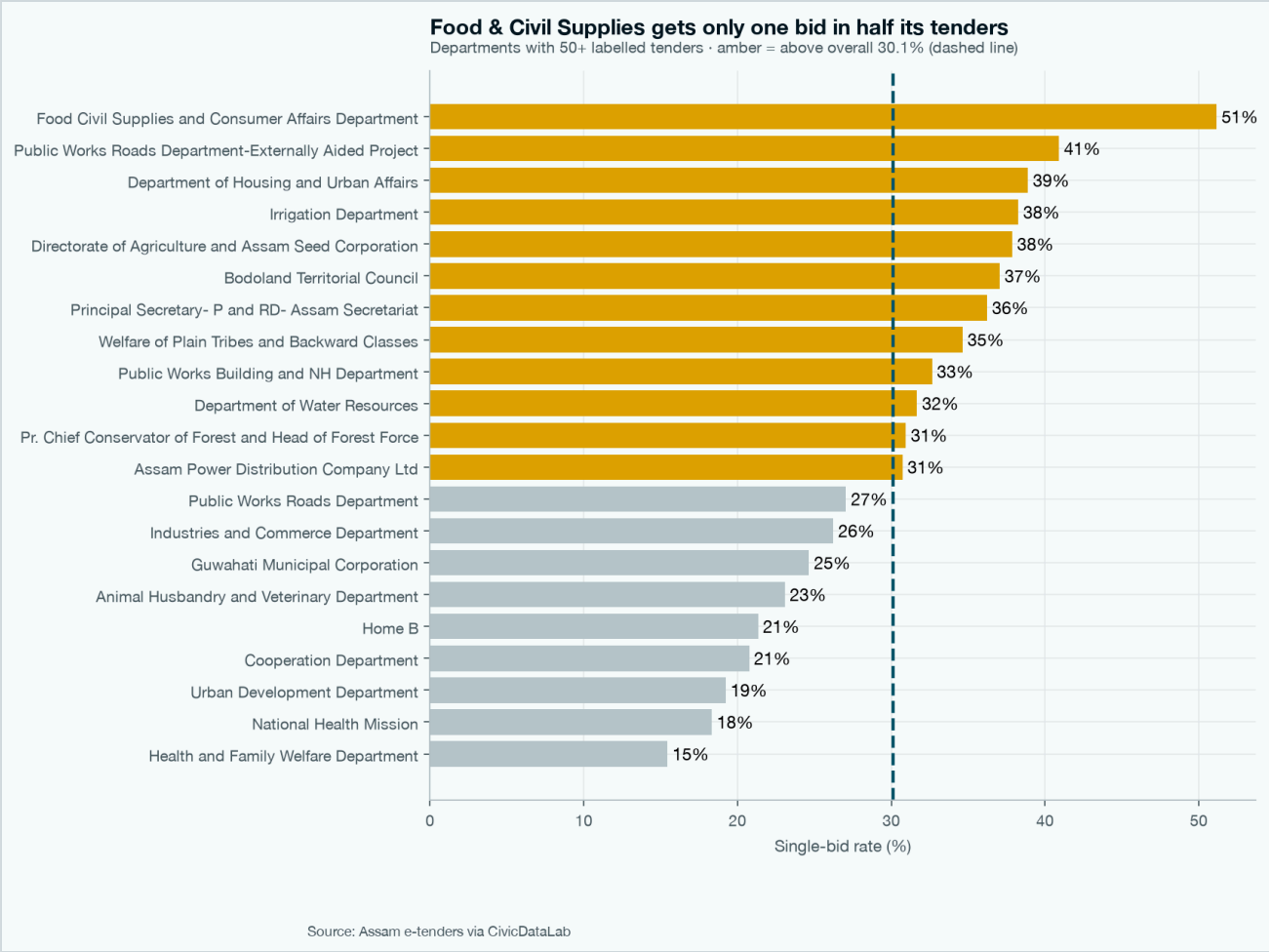


Figure 2 — Single-bid rate by department, departments with 50 or more labelled tenders

Three further patterns shaped the features built in the next stage. Tenders re-posted within 1 to 60 days of an earlier attempt had the highest single-bid rate of any group (40.0 percent). Tenders with no written value had a lower rate (23.3 percent) than those with a value (31.3 percent). And December 2022 stood out: 1,794 tenders were published in that one month, and the labelled ones had a single-bid rate of only 19.3 percent.

6.4 Feature Engineering and the Time Split

A model can only read numbers, and it must never see the answer or the future. Seven clues were therefore built using only information available on the day a tender is published:

Clue	Simple meaning
dept_rate	The department's past single-bid rate, smoothed by adding 20 "average" tenders so that small departments are not judged on a handful of cases
log_value	The tender value on a log scale, so that a Rs 2,000 crore project does not overpower normal tenders
value_missing	1 if the department did not write a value, otherwise 0
bid_window_days	Number of days firms are given to submit bids
months_to_fy_end	Months left until the financial year ends on 31 March (March = 0, December = 3, April = 11)

Clue	Simple meaning
same_day_lot	1 if the same department posted the same title on the same day (works split into lots)
retender_60d	1 if the same work was posted again within 1 to 60 days

Two leakage risks were handled carefully. First, the department rate for each training tender was calculated only from earlier tenders of the same department, so no tender could see its own answer inside its clue. Second, missing values were filled with medians calculated from the training data only.

The labelled tenders were then split by date rather than at random, because in real use a model always learns from the past and predicts the future. Tenders sharing an office reference number were kept together (46 shared numbers, 291 tenders moved forward), so that related tenders could not appear on both sides of the split.

Set	Period	Tenders	Single-bid rate	Used for
Train	Up to September 2022	2,753	36.1%	Learning
Validation	October to December 2022	2,270	27.8%	Choosing models and settings
Test	January to May 2023	1,767	23.8%	One final, untouched test



Figure 3 — The single-bid rate falls from training to validation to test

6.5 Model Building and Comparison

Every model was compared against two benchmarks, because a model is only useful if it beats simple alternatives:

- **Random checking:** what an auditor achieves by picking tenders at random.
- **Rule B:** a simple rule that ranks quick re-tenders first and then orders the rest by the department's past single-bid rate.

Logistic Regression, Random Forest and XGBoost were trained on the seven clues and on a wider set of 34 clues, using class weights to handle the imbalance between single-bid and multi-bid tenders.

Because only about one in four tenders is single-bid, accuracy would be misleading. The main measures were therefore PR-AUC (how well single-bid tenders are ranked near the top), ROC-AUC, and the share of single-bid tenders caught in the riskiest 20 percent, which matches how an auditor actually works.

On validation, no model beat random checking. Random checking scored a PR-AUC of 0.288, Rule B 0.268, Logistic Regression 0.252, Random Forest 0.233 and XGBoost 0.223. Adding more clues did not help. Rather than hiding this, the project investigated why, and found the most important result of the study: **drift**. Twenty-one of the 34 clues reversed their direction between training and validation, and a Random Forest could tell older tenders from newer ones with a ROC-AUC of 0.818. Departments that were risky before September 2022 often became safe afterwards, and the other way round (Figure 4).

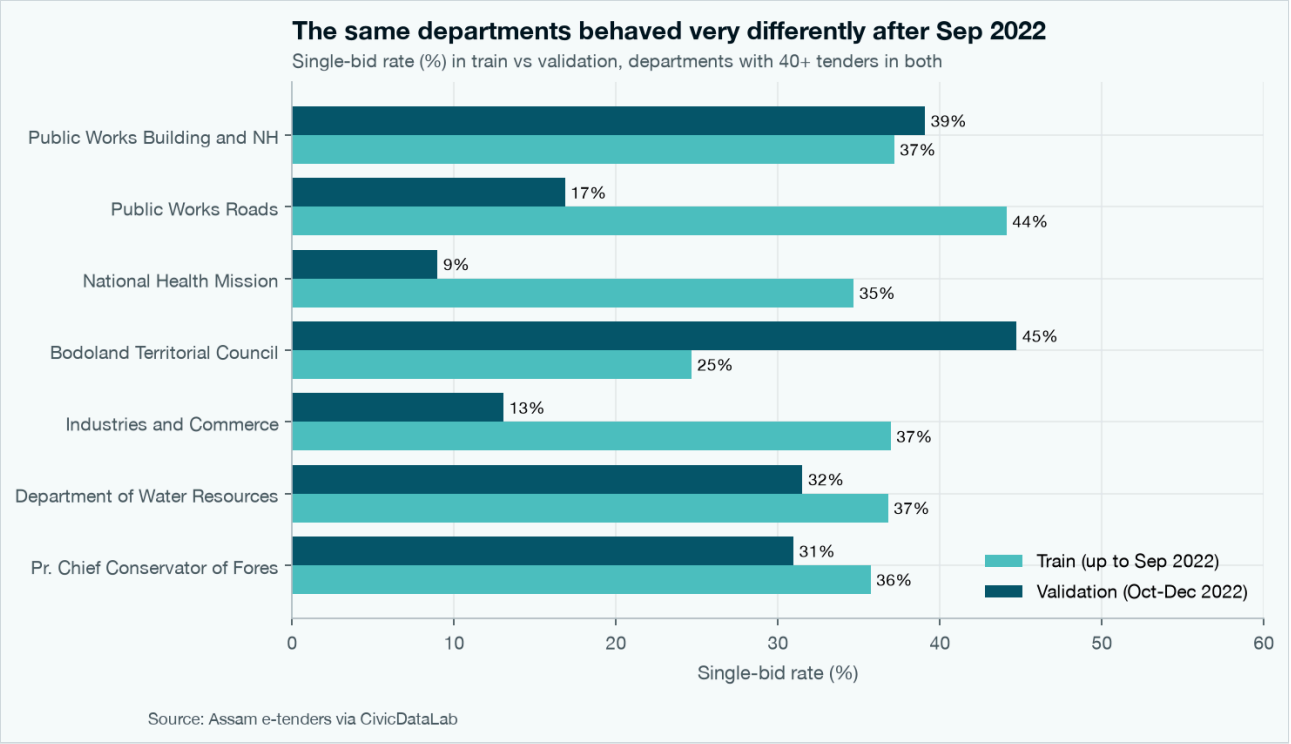


Figure 4 — The same departments behaved very differently after September 2022

6.6 Text Analysis (NLP)

The title and description of each tender were tested as an extra source of information, using two methods: TF-IDF, which counts important words, and sentence embeddings from the MiniLM model, which turn each text into 384 numbers that capture its meaning. Sentence embeddings were the best text method on validation, with a PR-AUC of 0.300, but this edge over random checking held in only 75 percent of 500 re-draws, which is not strong enough to call a real win. The text also confirmed the drift: a model could tell old tender wording from new wording with a ROC-AUC of 0.913, mainly because new government scheme names had replaced old ones.

This stage also uncovered a subtle form of leakage, called the **label timing** problem. When a model is retrained every month, it may only learn from tenders whose bids have already been opened, because an auditor cannot know the outcome of a tender that is still open. In validation, 1,098 of the 2,270 tenders opened their bids in a later month than they were published. Once this "opened only" rule was applied, every monthly retraining plan that had looked promising fell below random checking (Figure 5). The rule was adopted for the rest of the project.

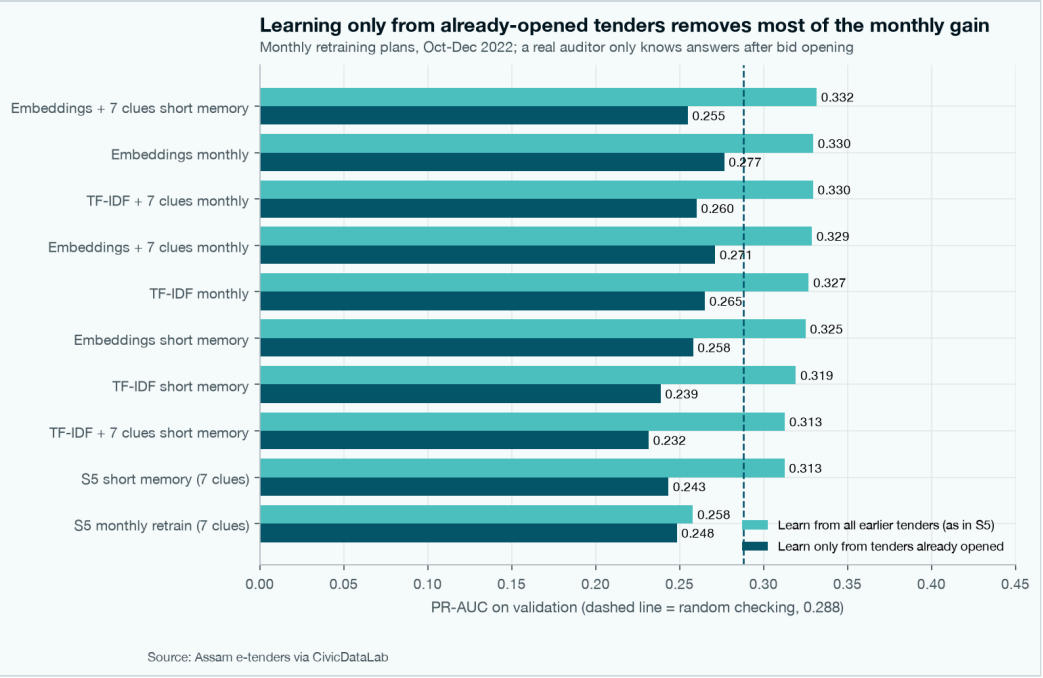


Figure 5 — Learning only from tenders whose bids were already opened removes most of the apparent monthly gain

6.7 The One Final Test

Before touching the test set, the complete plan was written down and saved with a timestamp (16 October 2026, 14:45). It fixed the methods to compare, the measures to report, the risk bands, and the rule for claiming a win. Four methods were tested: random checking, Rule B, the main model (Logistic Regression on the seven clues, retrained each month on already-opened tenders only), and sentence embeddings as a comparison. The test set was used exactly once.

A win was only claimed if it held up in a bootstrap test, in which the test tenders were re-sampled 500 times and the methods compared on each re-draw. This shows whether a difference is real or simply luck. The detailed results are presented in Section 11.

6.8 Anomaly Detection and Wider Patterns

An Isolation Forest was trained on ten publish-day clues to flag the most unusual 5 percent of tenders. It works by randomly splitting the data; tenders that can be separated from the rest in very few splits are unusual. For each flagged tender, the application shows which clues are the most different from a normal tender, so a reviewer knows why it was flagged. The stage also used the after-publication facts (bid counts, award prices, winning firms, amendments and re-posting) to study money at risk and the link between competition and price. These facts were analysed only, never used as model inputs.

6.9 Deep Learning

Finally, a neural network (ANN) was built on the same seven clues and the same split, so that the comparison with Logistic Regression would be fair. Two network sizes, with and without dropout, were tried, with early stopping on a separate part of the training data. A strict rule was set in advance: the neural network would only replace Logistic Regression if it beat both Logistic Regression and random checking in at least 95 percent of re-draws, and if every random starting seed also beat random checking. An autoencoder was also tested as an alternative to the Isolation Forest.

7. Data Architecture

The project is organised as a pipeline of files, where each stage reads the output of the previous one. Raw files are never edited; every change is made in code and saved as a new file. This makes the work reproducible: anyone with the raw zip file and the notebooks can rebuild every table, model and chart.

The pipeline moves through four layers. The raw layer holds the original zip file and the 39,500 tender CSVs. The interim layer holds the joined master table and the clean table. The processed layer holds the SQLite database, the feature tables with the time split, and the saved text embeddings. The model and application layer holds the trained models, the risk band cut-offs and the prepared app data that the Streamlit application reads.

File	Role	Rows
raw_data_fy_2016_2022.zip	Original download from CivicDataLab, one CSV per tender	39,500 files
raw_master.csv	All tender files joined into one table, before cleaning	39,500
tenders_clean.csv	Main clean table with 39 columns and the single-bid label	39,411
tenderlens.db	SQLite database used for the SQL exploration	39,411
features_split.csv	Labelled tenders with the seven clues and the train / validation / test split	6,790
features_split_v2.csv	Wider set of 34 clues used for comparison	6,790
s6_embeddings (two .npy files)	384 MiniLM meaning numbers for each tender's text	6,790
models/ folder	Final Logistic Regression, scaler, band cut-offs, Isolation Forest, and the comparison ANN	—
app/app_data/tenders.csv	Validation and test tenders scored for the app's review list	4,037
reports/ folder	About 60 result tables and 85 charts saved by the notebooks	—

This design has a practical benefit that shows up throughout the project: every number in this report and in the application can be traced back to a saved table, and the application never needs to repeat the heavy cleaning work. It simply loads the saved models and prepared data when it starts.

8. Data Description

Source: Assam e-tenders portal, published as open data by CivicDataLab (github.com/CivicDataLab/assam-tenders-data). **Coverage:** April 2016 to May 2023. **Unit:** one file equals one tender. Bidder, award and cancellation details are columns inside each tender's file, so all of this information comes from a single dataset.

Column group	Examples
Tender details	Tender ID, reference number, department, title, description, category (works, goods, services)
Money	Estimated tender value, EMD (earnest money deposit) amount, tender fee
Dates	Published date, last date for bids, bid opening date, work period in days
Bidders	Full list of bidders (newer files only), bidder status
Award	Winning firm, awarded value
Other events	Cancellations, amendments (corrigenda), location text

Figure 6 shows how the data narrows from raw files to the labelled modelling set. Only about one in six tenders has the bidder list the model needs. Of the labelled tenders, 3,610 also have award details, of which 2,933 give a clean award ratio, and 1,249 tenders were cancelled before their bids were opened.

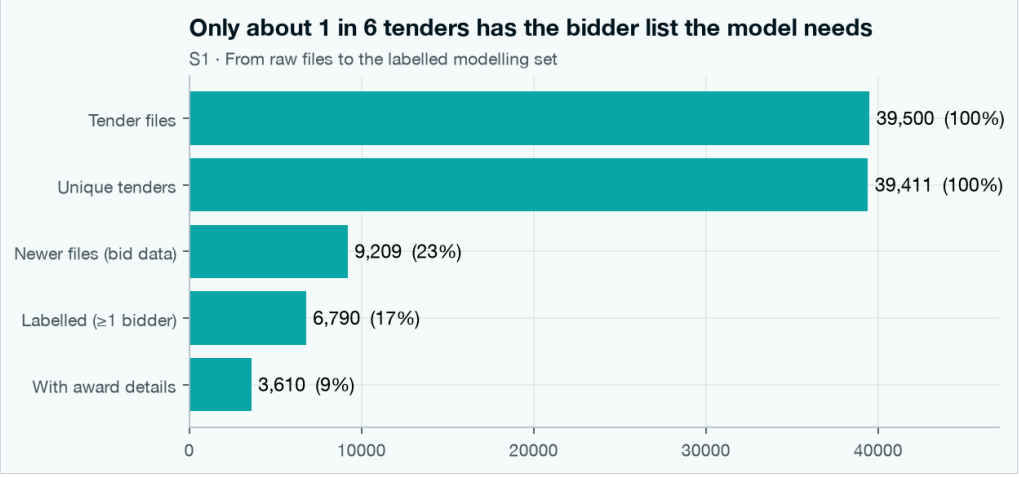


Figure 6 — The data funnel: 39,500 files, 39,411 unique tenders, 6,790 labelled tenders

The number of tenders per year also varies: 571 in 2016, 4,961 in 2017, 9,318 in 2018, 6,004 in 2019, 6,723 in 2020, 3,060 in 2021, 6,293 in 2022 and 2,570 in January to May 2023. Most labelled tenders come from 2022, with only about 330 before that year.

9. Application Design

The results are delivered through TenderLens, an eight-page Streamlit application. It was designed around one question that a reviewer asks every month: *which tenders should I check first, and why?* The app uses a navy sidebar for navigation, a warm cream background, and the same colour language as the analysis charts: amber for single-bid risk, teal for competition, and grey for routine tenders. It loads the final Logistic Regression model and the Isolation Forest from the saved model files, and it scores the 4,037 validation and test tenders, ranking each month's tenders on their own. Figure 7 summarises how it works.

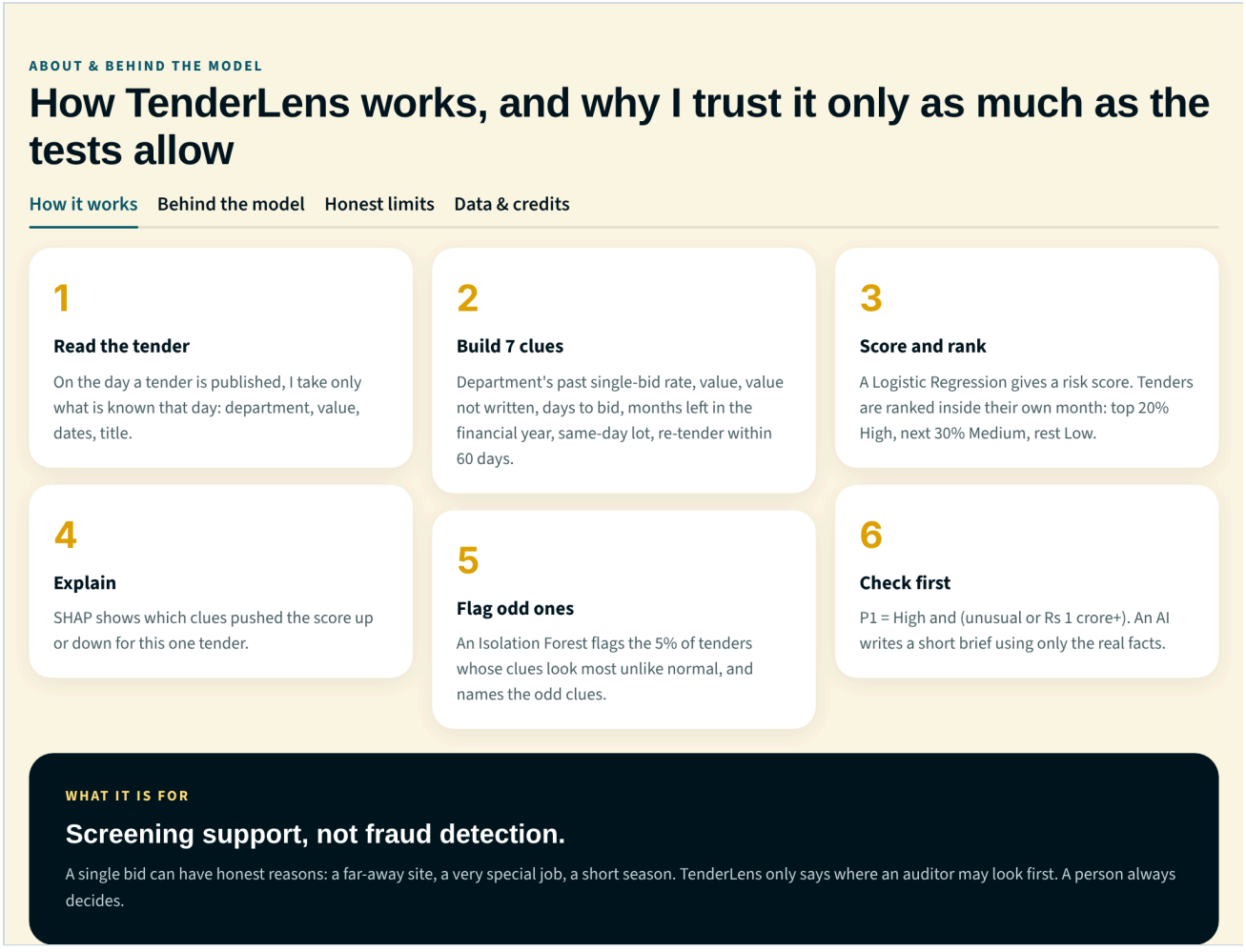


Figure 7 — How TenderLens works: the six steps behind every page (from the About page)

Three design decisions run through every page. First, risk is shown as a band rather than a percentage: the riskiest 20 percent of a month's tenders are High, the next 30 percent Medium, and the bottom 50 percent Low. This is because the model's scores rank tenders well but are not true probabilities. Second, the single-bid risk band and the unusual flag are shown as two separate signals. Third, a simple **check-first rule** combines them: Priority 1 is a High-band tender that is also unusual or worth at least Rs 1 crore; Priority 2 is High only or unusual only; everything else is routine.

9.1 Overview

The opening page answers the question "where should an auditor look first this month?" A month selector runs across the top, from October 2022 to April–May 2023. A dark "why this matters" panel

states the headline money finding, Rs 14,752 crore of tender money that went out with only one bidder, alongside four key numbers: 36.5 percent of the money, four or more bidders for real discounts, 101 against 79 caught compared with random checking, and one tender in seven to check first. Below it, five cards summarise the selected month. For January 2023, these show 842 tenders, 168 in the High band, 13 unusual tenders, 109 Priority 1 tenders and a real single-bid rate of 28.5 percent. Two charts show the falling single-bid rate after September 2022 and the risk bands working in the right order on the fair test. Clicking a bar opens that month's list.



Figure 8 — Overview page: headline finding, monthly summary cards and the bands chart

9.2 Tender Risk List

This is the main working page. All of a month's tenders are ranked from most to least risky inside that month, with their band, check-first level and signals (unusual, re-tender, data alert). An **audit budget** slider lets the reviewer say how many tenders they can check. The page then replays the month fairly, using the scores the model gave before the bids opened, and shows how many single-bid tenders that budget would have found compared with random checking. With 168 checks in January 2023, following the TenderLens order finds 60 single-bid tenders against an average of 48 by random checking, 12 more, covering Rs 303 crore of tender value. Filters for search text, band and department, quick toggles for Priority 1, unusual, data-alert and re-tender tenders, and a CSV download of the full list or the check-first list are provided. Clicking a row opens the tender's detail page.

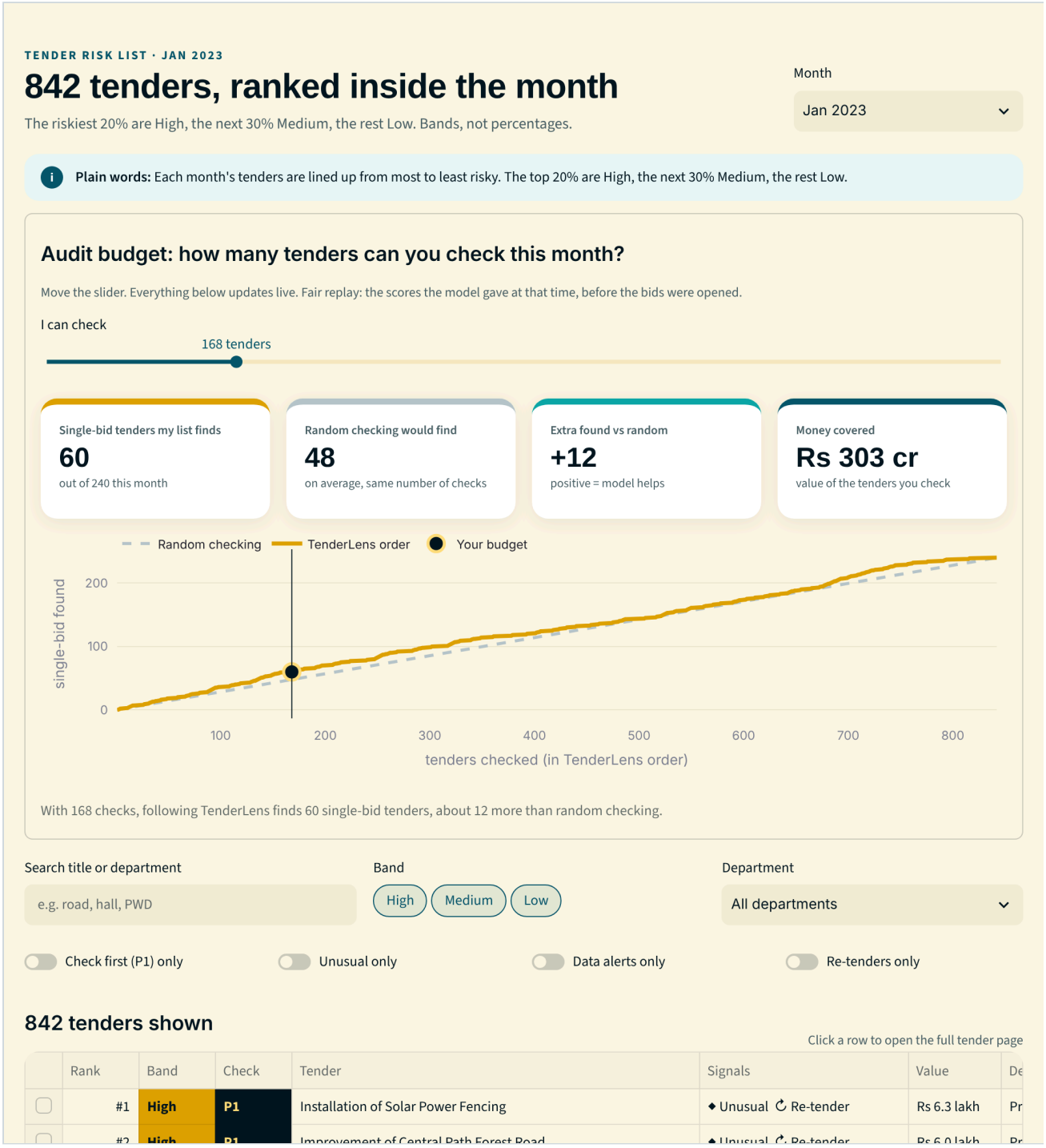


Figure 9 — Audit budget: 168 checks find 60 single-bid tenders, 12 more than random checking



Figure 10 — The ranked list with bands, check-first levels, signals and download buttons

9.3 Tender Detail and the GenAI Review Brief

The detail page explains one tender completely. The example shown is "Installation of Solar Power Fencing", ranked first of 842 tenders in January 2023 and marked High band, Priority 1, unusual and re-tender. A risk meter shows its position within the month, a key facts card lists everything known on the day of publication, and a re-tender history box shows that the same title was posted and cancelled 14 days earlier. A "possible saving" card gives a rough sign of what more competition could save, worded clearly as a sign and not a promise. The reviewer can mark the tender as checked, add a note, and download a one-page PDF.

Tab	What it shows
Why	SHAP bars: each clue that pushed this tender's risk up (amber) or down (teal)
Unusual	The Isolation Forest score and the two clues that are furthest from a normal tender
Similar	Earlier tenders with similar titles, and how many bids each one received
What-if	Change the bidding window or the value and see how the model's view changes (not proof that more firms would bid)
Review brief	A short, fact-checked plain-English brief written by an LLM, described below

Five tabs go deeper, as summarised below. For this tender, the Why tab (Figure 12) shows that being re-tendered within 60 days pushed the risk up the most.

The possible saving card is calculated simply and openly. Across all awarded tenders, the median award was 96.1 percent of the estimate with one bidder and 90.0 percent with four or more bidders. The card multiplies the tender value by this 6.1 point gap; for this Rs 6.3 lakh tender it gives about Rs 38,590. It is shown as a sign of what more competition could be worth, not as a promise or a finding of loss.

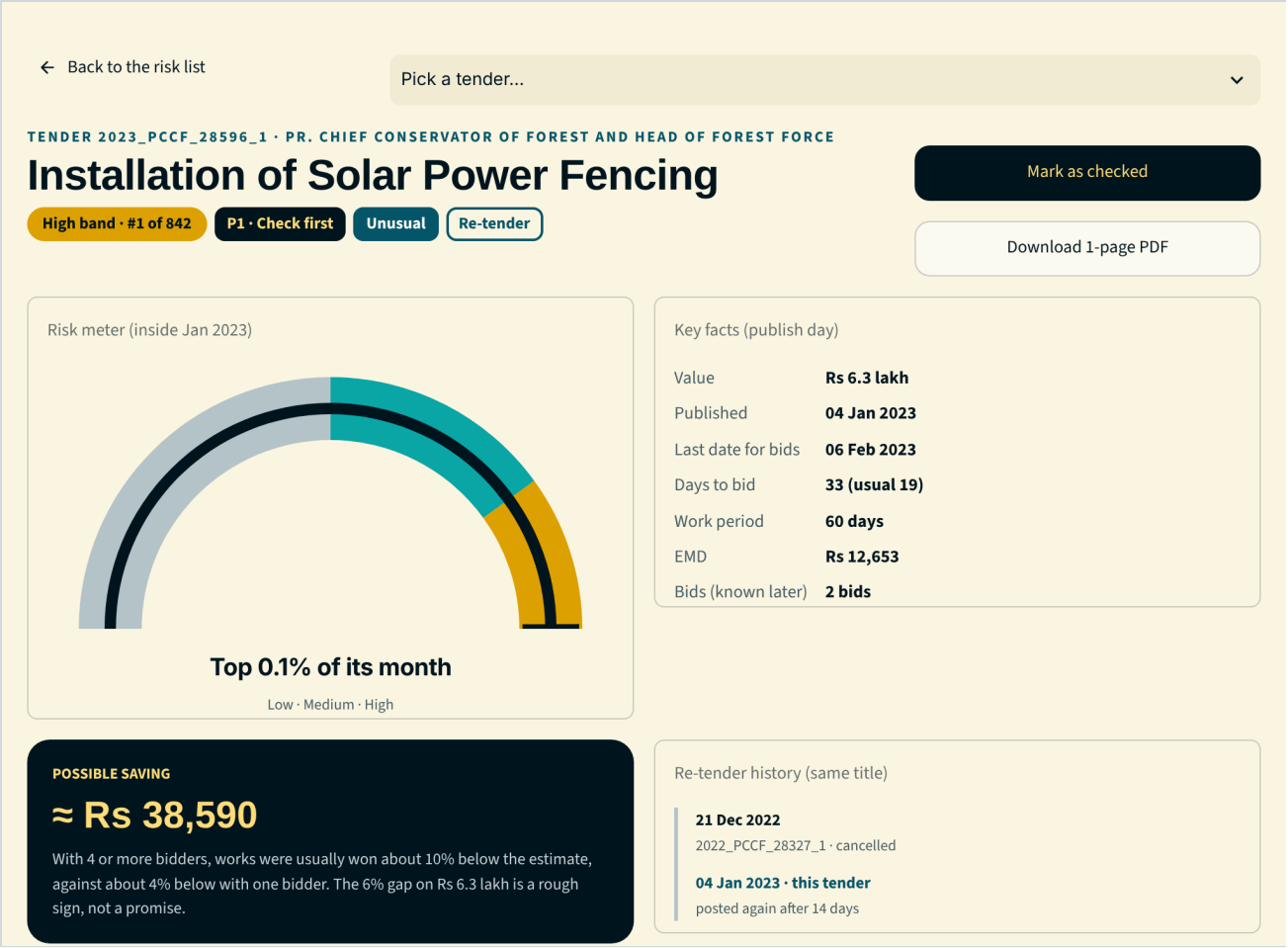


Figure 11 — Tender detail: risk meter, key facts, possible saving and re-tender history

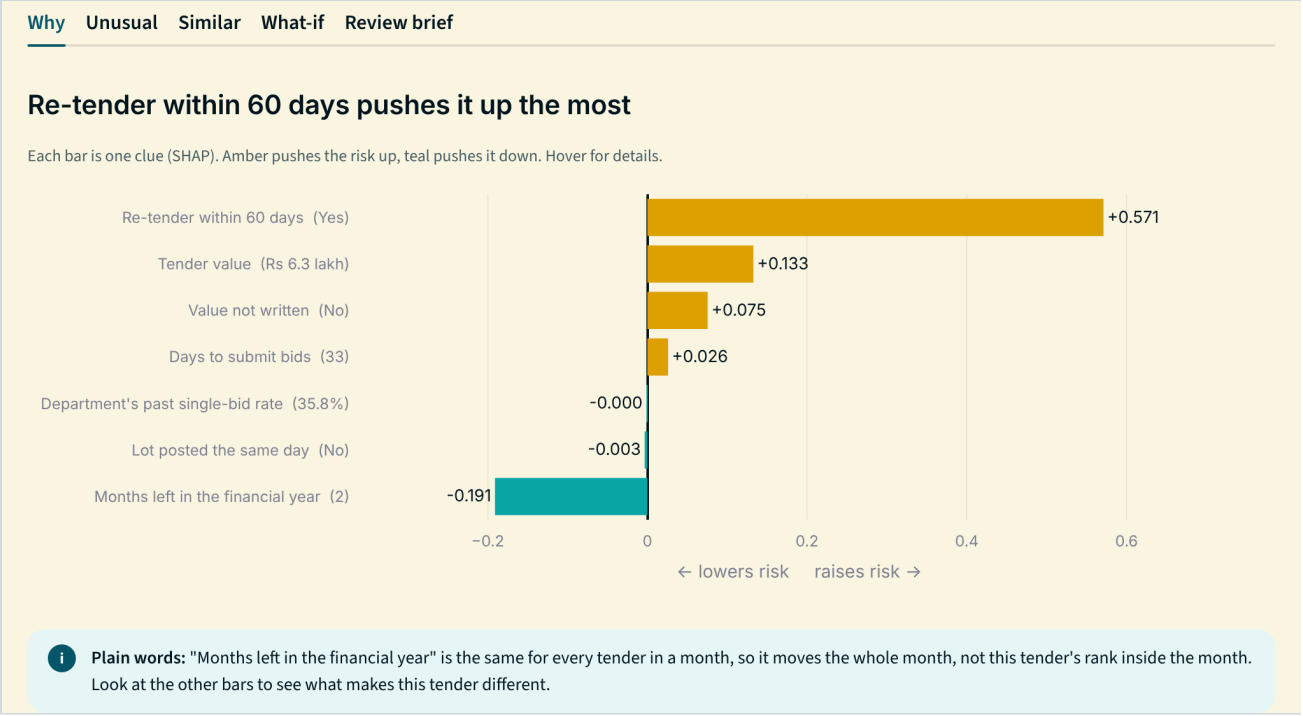


Figure 12 — Why tab: SHAP bars show which clues pushed this tender's risk up or down

The **Review brief** tab is the project's generative AI feature. With one click, a large language model (Google Gemini or Groq, set through a private key) writes a four to five line plain-English brief for the

tender. The model is given only the tender's verified facts, its SHAP reasons and its unusual clues, which are shown next to the brief so the reviewer can see exactly what the AI was told. After writing, every number in the brief is automatically checked against these facts. If the AI service is unavailable, or the check fails, a rule-based template writes the brief instead, so the feature never stops working and never invents facts. A Hindi version of the template is also available. The example below was written by the template and passed the fact check: all ten numbers were found in the facts.

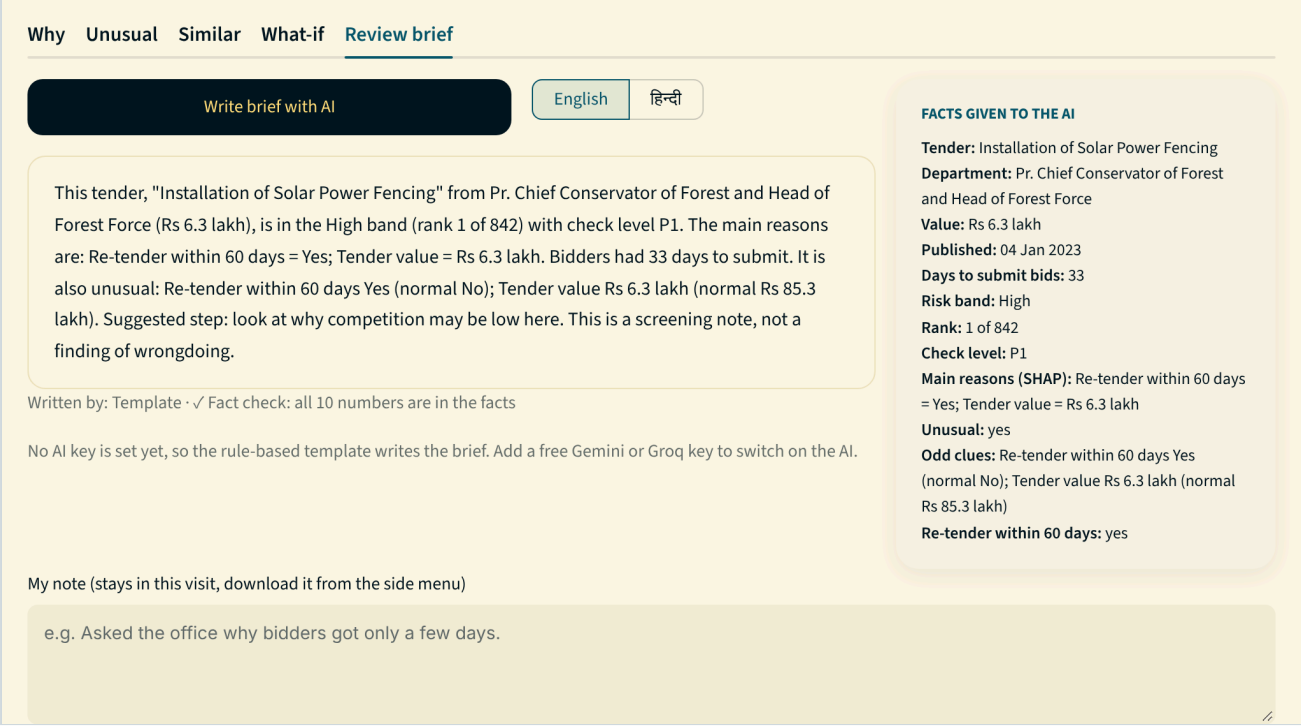


Figure 13 — Review brief: a short, fact-checked brief with the facts given to the AI shown alongside

9.4 Department Report Card

This page gives one department's full picture. The example below is Assam Power Distribution Company Ltd, the department with the most single-bid money. Supplier names are replaced by anonymous codes (S1, S2 and so on), so the page never blames a firm by name. A quarter-by-quarter chart shows the department's trend, a dot chart compares every department before and after September 2022 (clicking a dot switches to that department), and the page ends with the department's current riskiest tenders.

Card on the page	Assam Power Distribution Company Ltd
Labelled tenders	231
Single-bid rate (all years)	30.7%
Single-bid money	Rs 4,712 crore (31.9% of all single-bid money)
Single-bid rate before and after September 2022	35% before, 18% after
Awards won by repeat winners (3 or more wins)	69%
Share of awards won by the top firm	9%

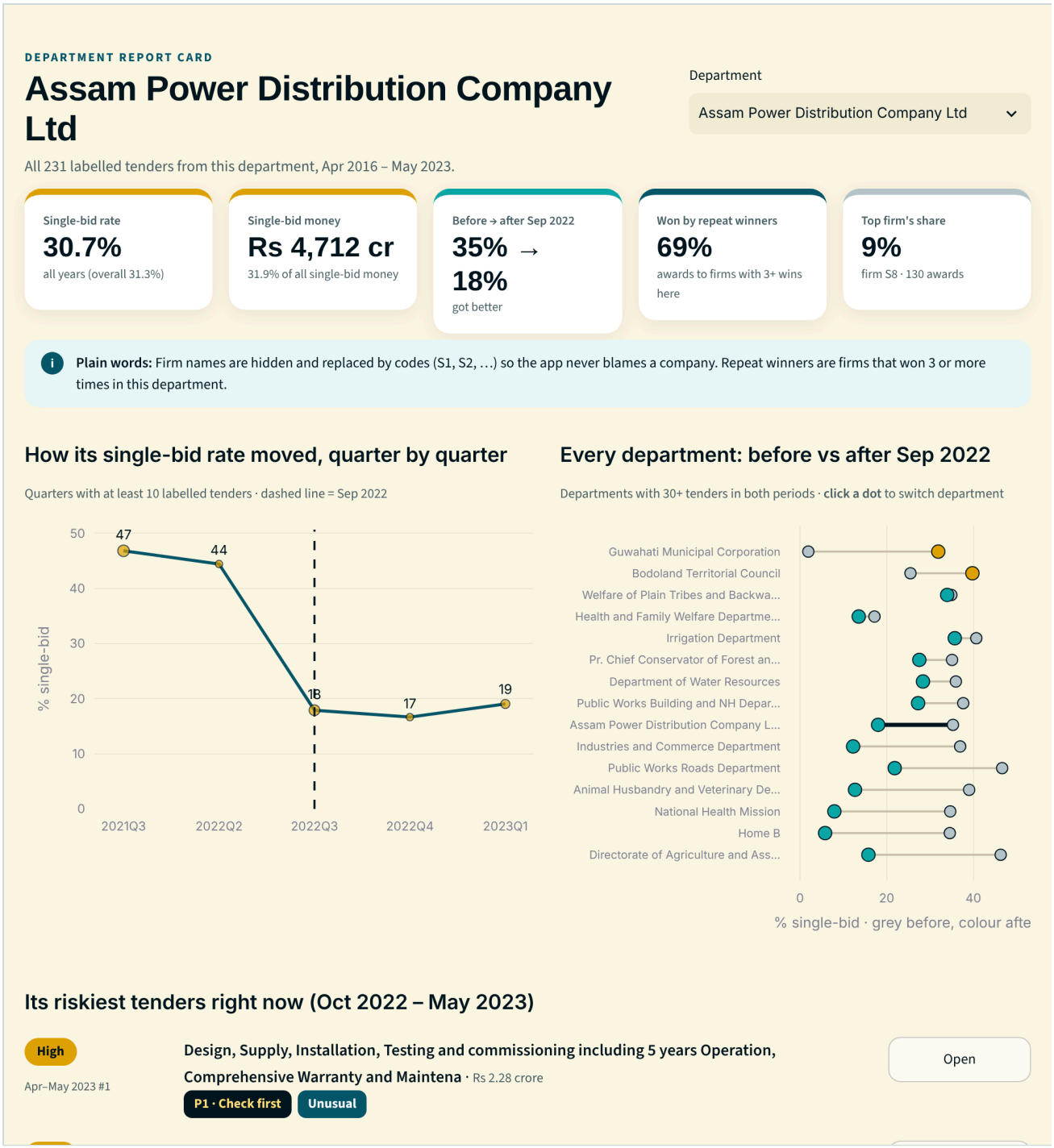


Figure 14 — Department report card for Assam Power Distribution Company Ltd

9.5 Model Health

Because the data drifts, the app checks itself every month. The first check is a fair replay: how many of the month's single-bid tenders fell in the riskiest 20 percent, compared with the 20 percent expected from random checking. Across October 2022 to May 2023, the model caught 227 of 1,051 single-bid tenders (21.6 percent) and beat random in four of seven months. The second check measures drift with the Population Stability Index (PSI), which compares each month's tenders with the training data: below 0.1 means little change, 0.1 to 0.25 some change, and above 0.25 a warning to check the model. Drift was strongest in November and December 2022 (0.28 and 0.30) and in April–May 2023 (0.51), where the bidding window changed the most. A heat map shows which clue changed and when.

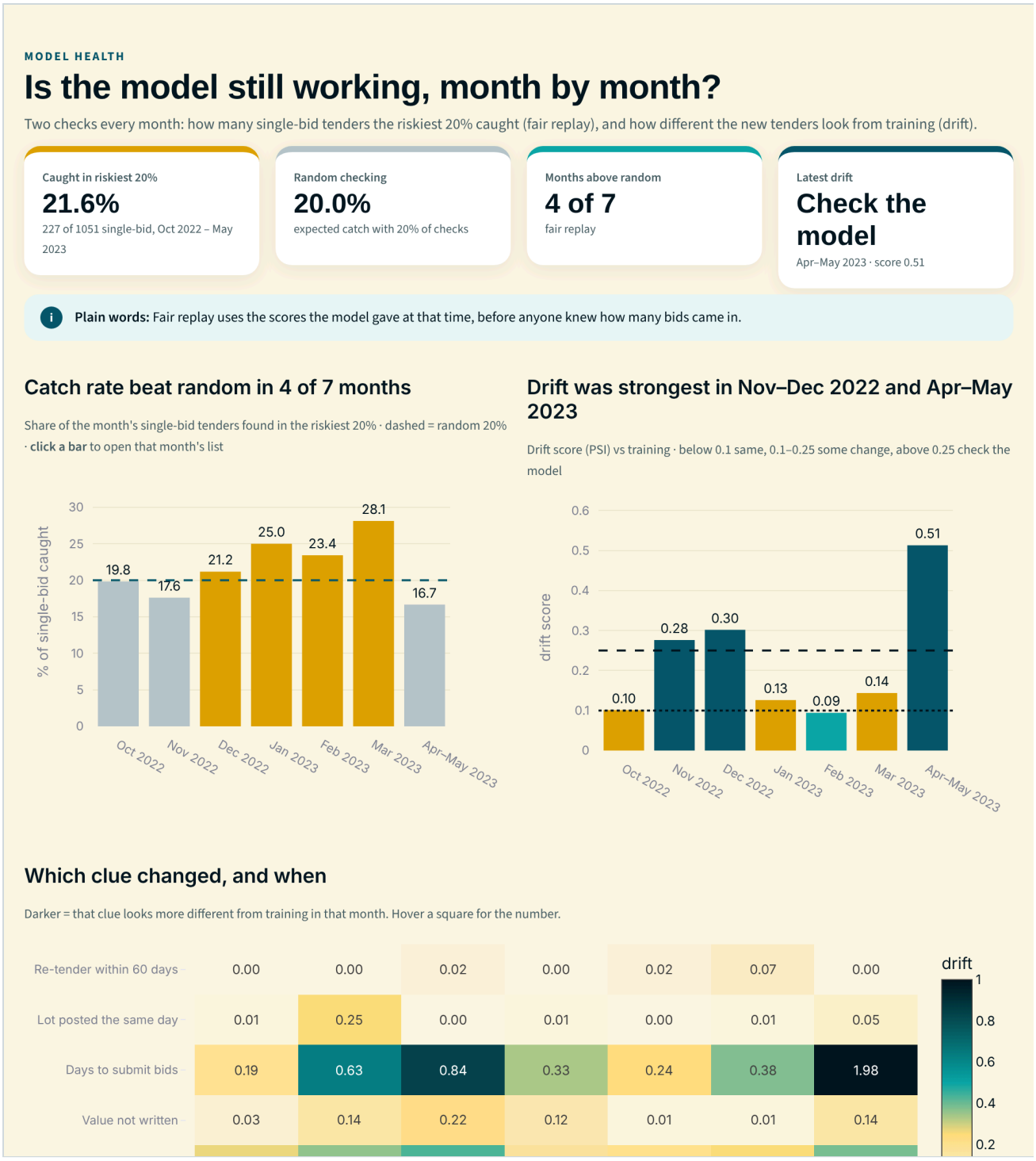


Figure 15 — Model health: monthly catch rate against random, and the drift score for each month

9.6 Score New Tenders

This page shows how TenderLens would be used on tenders that have just been published. A reviewer can either type in one tender through a short form and press Predict, which opens the full detail page for it, or upload a CSV file of many tenders using a provided template. The app validates the columns, builds the seven clues exactly as in the feature engineering notebook, fills any blanks with the training medians, scores each tender with the saved model and places it in a band by comparing it with tenders from the same month of the financial year. With the sample file of 40 real tenders from April–May 2023, the app placed 9 in the High band, flagged 6 as unusual and marked 2 as Priority 1. The scored file can be downloaded.

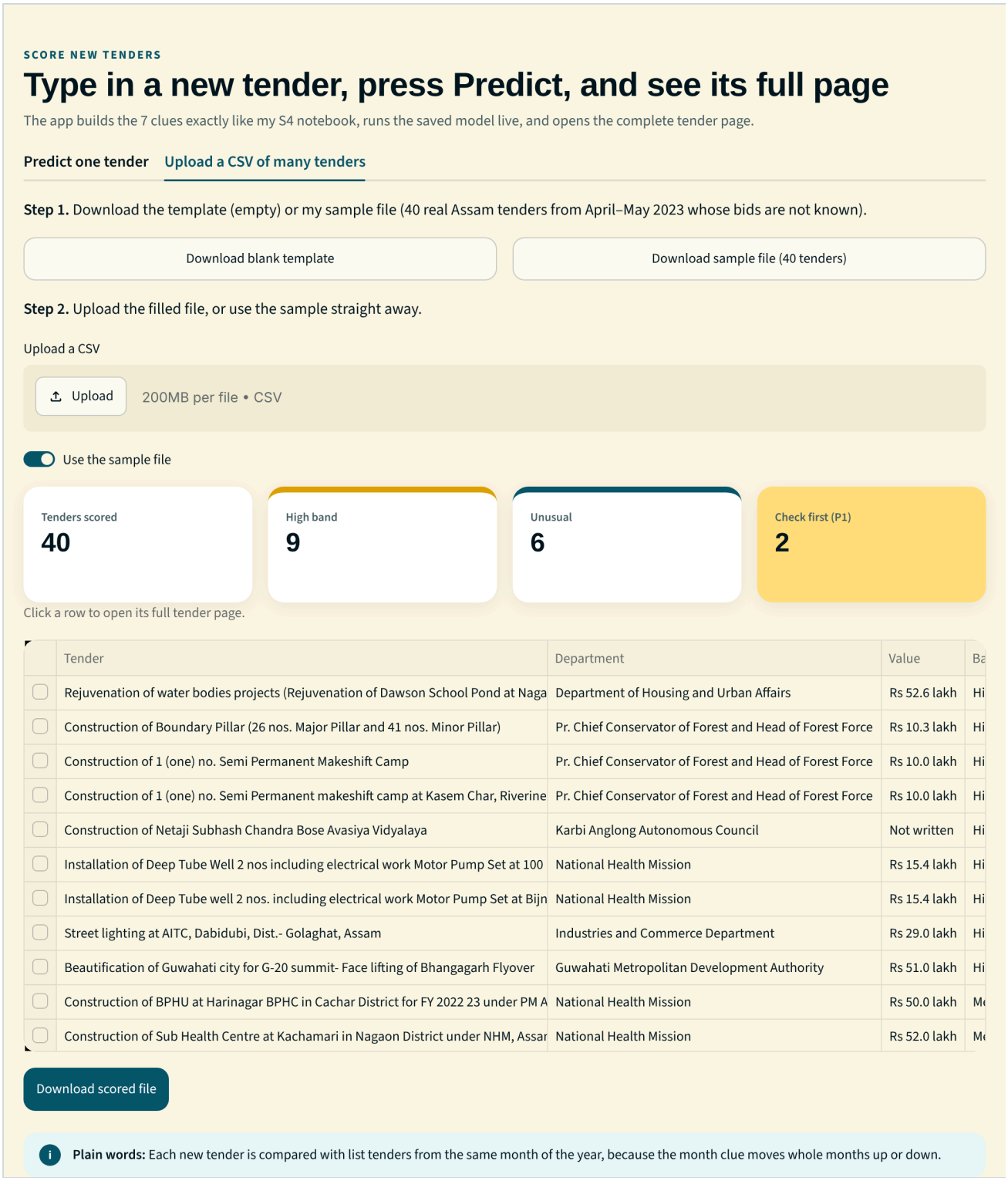


Figure 16 — Score new tenders: the sample upload of 40 tenders, scored and banded

9.7 Beat the Model

This interactive page turns the evaluation into a game. The user sees 30 real tenders from a past month with only their publish-day facts, picks the ten they would check, and then compares how many single-bid tenders they found against TenderLens's top ten and ten random picks. It is a simple way to show an audience that a screening tool helps, while also showing honestly that small rounds are noisy and that the model is only a modest improvement on careful judgement.

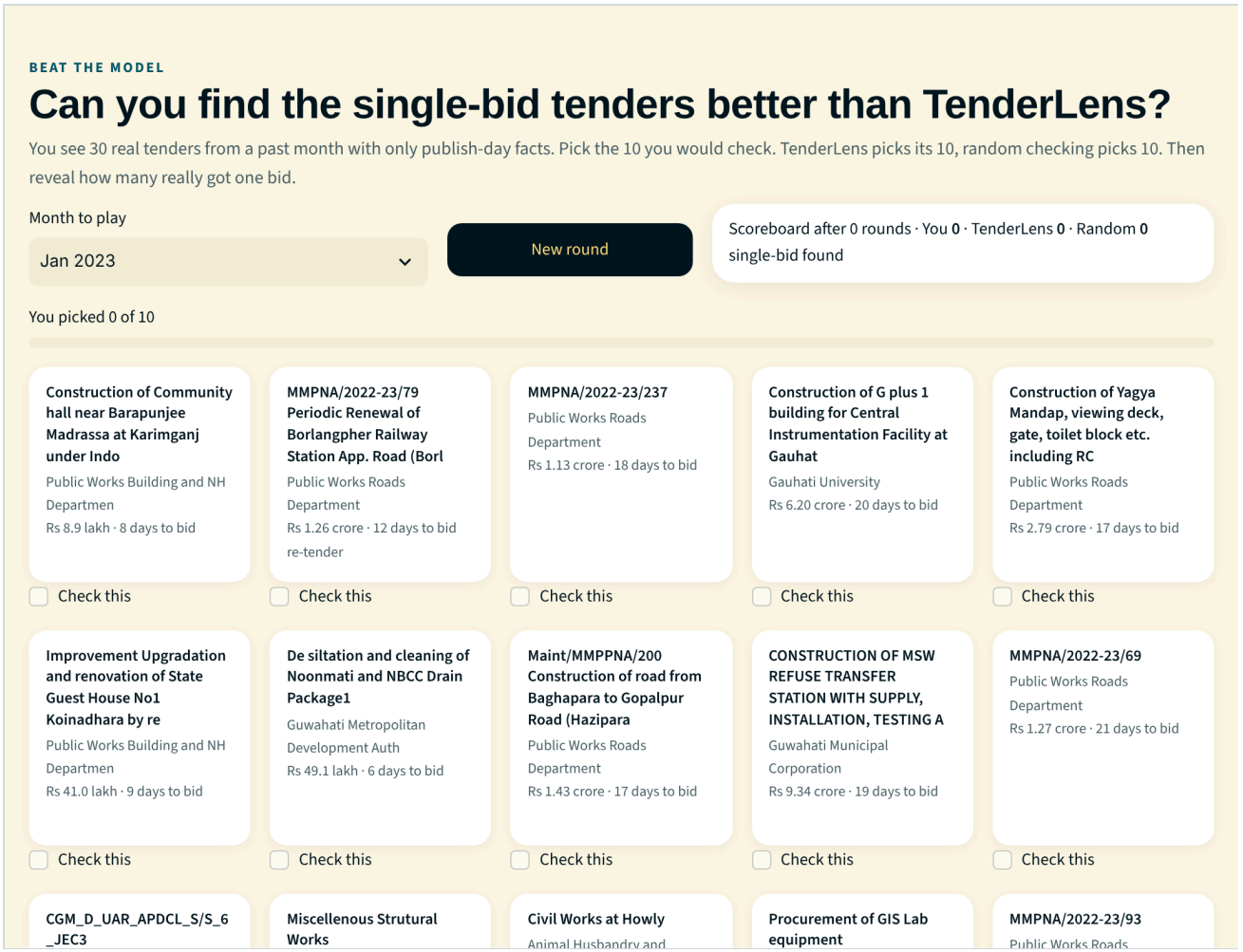


Figure 17 — Beat the model: pick ten tenders and compare with TenderLens and random checking

9.8 About and Behind the Model

The final page explains how TenderLens works in six simple steps (read the tender, build seven clues, score and rank, explain, flag odd ones, check first), shown in Figure 7 at the start of this section, followed by tabs on the model's evidence, its honest limits, and data credits. It states clearly that the tool is screening support, not fraud detection, and that a person always makes the final decision. A plain-words switch in the sidebar adds simple explanations throughout the app for non-technical users.

10. Validation and Quality Assurance

A model is only as trustworthy as the way it is tested, so validation was treated as a required step at every stage rather than an afterthought. The biggest danger in a project like this is **data leakage**: the model accidentally seeing information it would not have in real life, which makes results look far better than they really are. The following safeguards were built in:

- **Publish-day inputs only.** Bid counts, bidder statuses, award values, winning firms, cancellations, amendments and re-posting results were never used as inputs.
- **Time-based split.** The model always learned from the past and was tested on the future, never on a random mix of both.
- **Related tenders kept together.** Tenders sharing an office reference number were kept in the same set.
- **Earlier-only department rates.** Each training tender's department rate used only earlier tenders, so no tender saw its own answer.
- **Training-only fill values.** Missing values were filled with medians from the training data only.
- **Label timing check.** Monthly retraining learned only from tenders whose bids had already been opened. This check removed a gain that would otherwise have been reported as a success.
- **Locked plan.** The final test plan was saved with a timestamp before the test set was opened, and the test set was used once.

Results were also checked for stability. Every claimed win was tested with 500 bootstrap re-draws. The neural network and the Isolation Forest were re-run with several random seeds; the Isolation Forest flagged between 73 and 85 test tenders across seeds, with 84 to 93 percent agreement, while the autoencoder agreed only 17 percent of the time for two of five seeds and was therefore not used.

Finally, the numbers themselves were cross-checked. Every notebook was first run in a separate environment and then re-run on the project laptop; the two runs matched exactly, apart from tiny differences in the sixth decimal place. The application's department money totals were checked against the analysis notebook and matched (Rs 14,752 crore in single-bid tenders). Inside the application, every GenAI brief is automatically checked so that any number in the text must match a verified fact, and a template brief is used if the check or the AI service fails.

> *The project's central rule was simple: a result is only reported as a win if it survives a fair test on future data. Several promising results did not survive, and they are reported as such.*

11. Key Insights

This section draws together the project's most important findings in one place. Every figure below comes from a saved result table in the project, produced by the notebooks described in Section 6.

11.1 Single-bid tenders carry more than their share of public money

Of the 5,771 labelled tenders with a written value, worth Rs 40,450 crore in total, single-bid tenders accounted for 31.3 percent of tenders but 36.5 percent of the value: about Rs 14,752 crore. Single-bid works tend to be larger. The money is also highly concentrated: three departments hold 75.3 percent of all single-bid value — Assam Power Distribution Company (Rs 4,712 crore), PWD Building and National Highways (Rs 4,574 crore) and PWD Roads, Externally Aided Projects (Rs 1,836 crore).

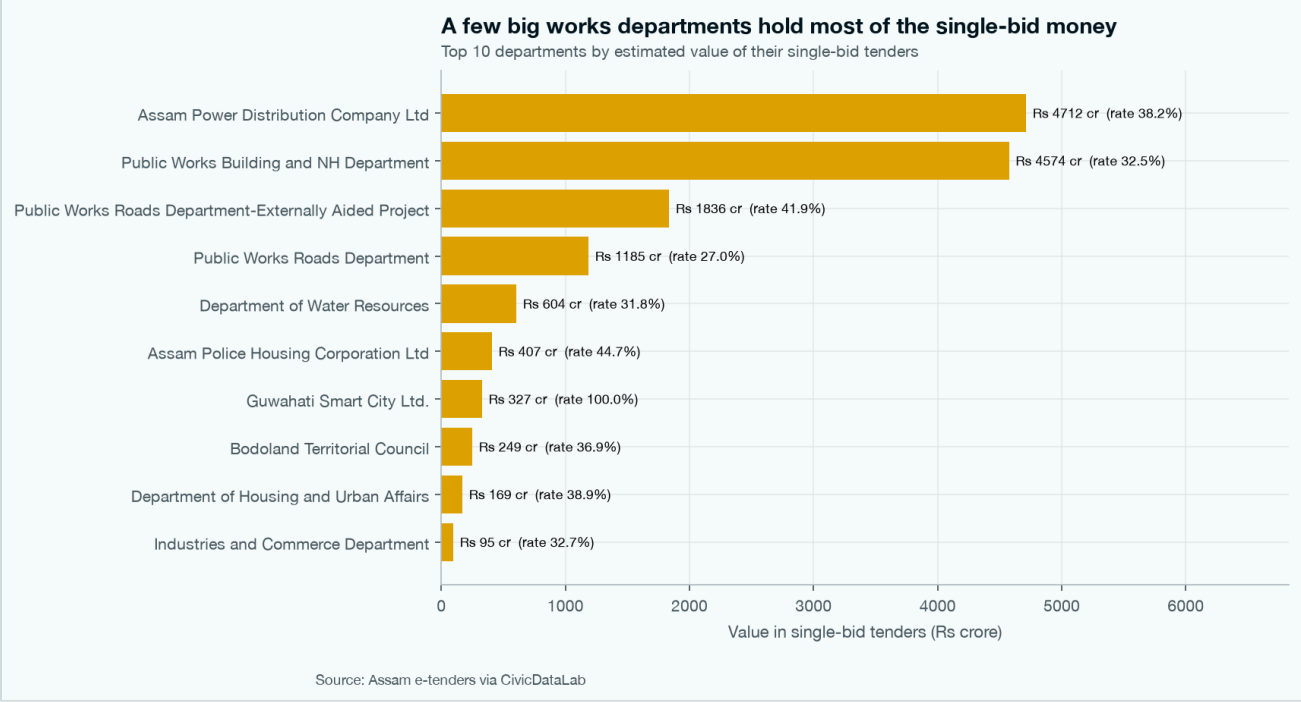


Figure 18 — Single-bid money by department: three departments hold three quarters of the value

11.2 Real discounts only appear with four or more bidders

Number of bids	Awards	Median award / estimate	Awarded exactly at the estimate
1 bid	1,481	0.961	32.1%
2 bids	397	0.999	38.3%
3 bids	477	1.000	53.0%
4 bids	209	0.900	17.2%
5 or more bids	369	0.900	4.3%

Comparing the awarded price with the estimate shows how much competition matters. With one bid, the median award was 96.1 percent of the estimate. With two or three bids, the price stayed at the estimate (99.9 and 100 percent), and over half of three-bid tenders were awarded exactly at the esti-

mate. Only with four or more bids did the median award fall to 90 percent. A rough calculation puts the cost of single bidding at Rs 120 to 680 crore, depending on the comparison used. This is a sign, not proof, because single-bid tenders may differ in other ways.

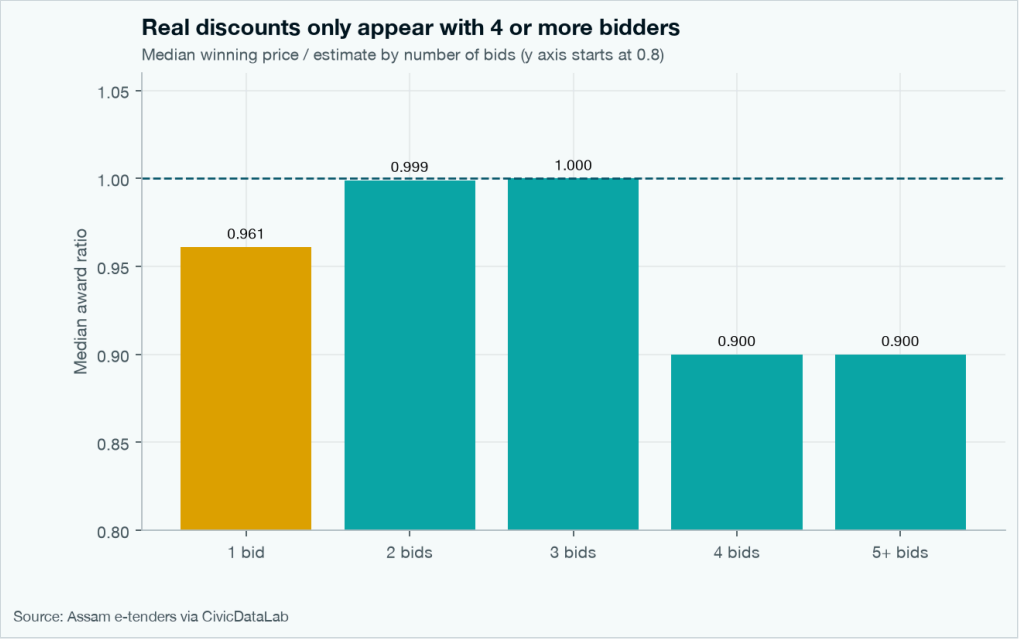


Figure 19 — Median award as a share of the estimate, by number of bids

11.3 Tender patterns changed sharply after September 2022

The single-bid rate fell from 36.1 percent in training to 27.8 percent in validation and 23.8 percent in test. This was not a simple shift: 21 of 34 clues reversed direction, departments changed places, and new scheme names replaced old ones in the tender text. One-bid tenders moved mostly into two and three bids. Thirteen of fifteen large departments improved, with Agriculture falling from 46.3 to 15.8 percent and PWD Roads from 44.1 to 22.9 percent. This drift is the main reason modest model scores are expected, and it is why the project tests on future data only.

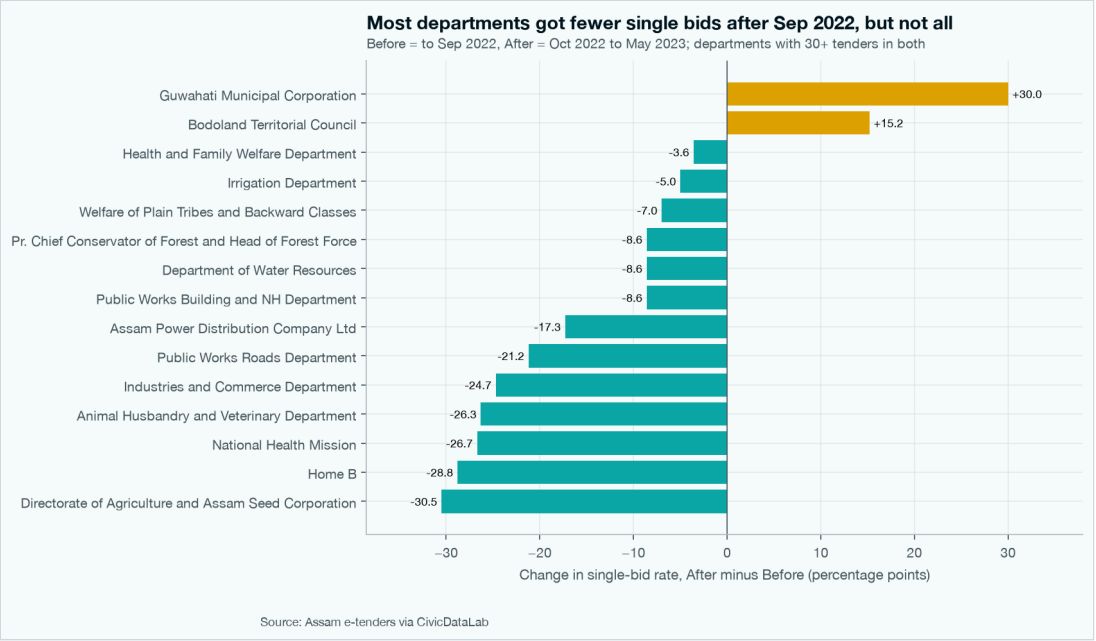


Figure 20 — Thirteen of fifteen large departments improved after September 2022; two got worse

11.4 The final model beats random checking, by a small but real margin

On the one final test (January to May 2023, 1,767 tenders, 421 single-bid), an auditor checking the model's riskiest 20 percent (353 tenders) would have found 101 single-bid tenders, against 79 with random checking: 22 more tenders found for the same effort, a lift of 1.20. The model won in 95.8 percent of 500 re-draws. Rule B found 108, so a simple rule remains a strong baseline; it is reported openly, and it is the reason the application combines the model with re-tender information rather than relying on the score alone.

Method (test, January–May 2023)	PR-AUC	ROC-AUC	Caught in top 20%	Capture
Random checking	0.232	0.499	79	18.8%
Rule B: re-tender, then department rate	0.279	0.507	108	25.7%
Main model: Logistic Regression, 7 clues, monthly	**0.256**	**0.552**	**101**	**24.0%**
Sentence embeddings (comparison)	0.245	0.507	85	20.2%



Figure 21 — Checking the riskiest 20 percent: main model 101, Rule B 108, random 79

The risk bands also worked in the right order on test: High-band tenders were single-bid 28.6 percent of the time, Medium 24.9 percent and Low 21.3 percent, as shown in the bands chart on the application overview page (Figure 8). In money terms, however, the model did not find more value than random checking (Rs 170 crore against Rs 200 crore), so an earlier idea that it favours bigger tenders was not confirmed.

11.5 Timing and a missing value are the strongest clues

SHAP shows how much each clue pushes a tender's risk up or down. The two strongest clues were the number of months to the end of the financial year and whether the value was left empty. Tenders published early in the financial year (April and May) were pushed towards higher risk, while those with no written value were pushed lower, because empty values are common in goods and supply tenders from certain offices, which in this data receive a single bid less often. Because the timing

clue is the same for every tender in a month, it shifts whole months up or down; this is why the application ranks each month's tenders against each other.

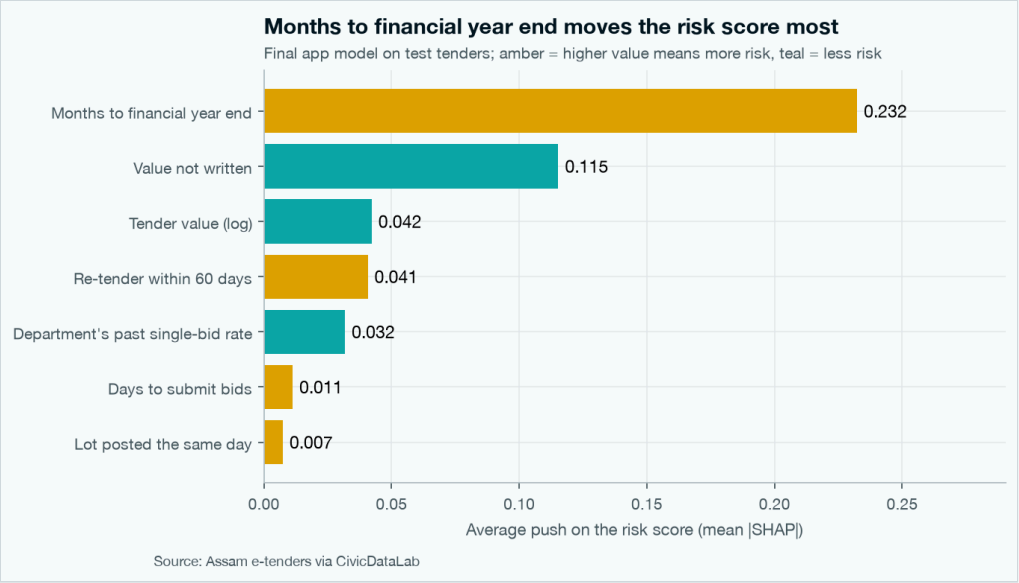


Figure 22 — SHAP importance: months to year-end and value missing matter most

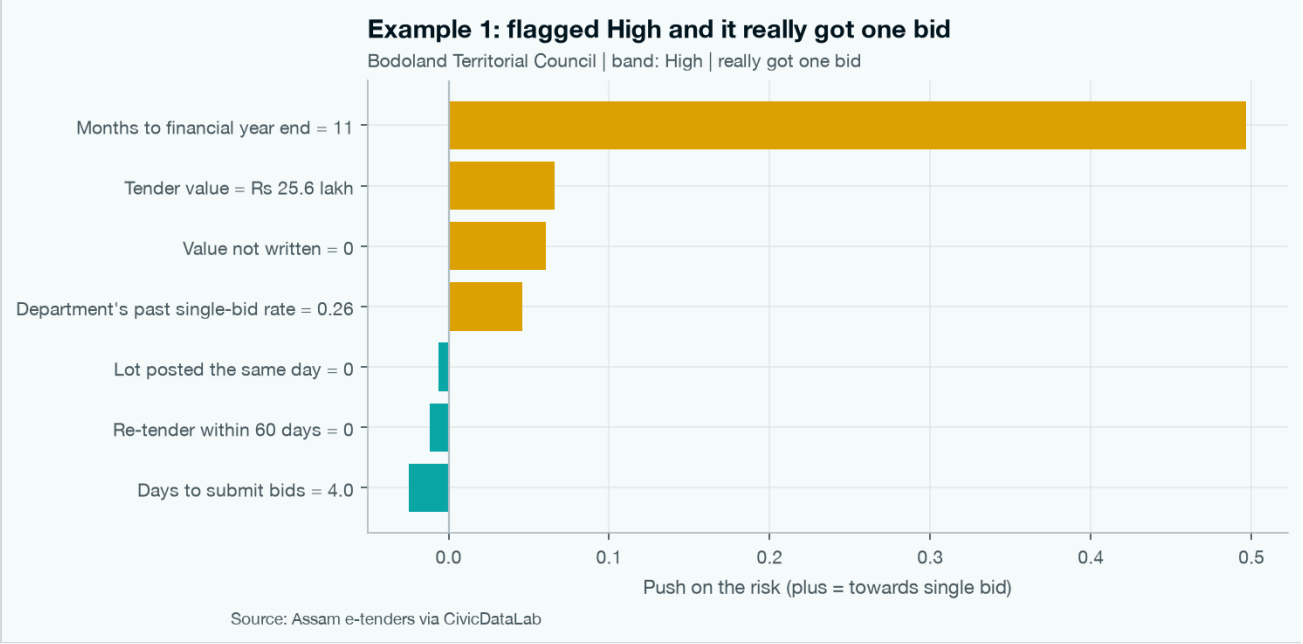


Figure 23 — SHAP explanation for one correctly caught single-bid tender

11.6 A simpler model was kept because the neural network did not earn its place

On validation, the neural network scored a PR-AUC of 0.286, essentially the same as random checking (0.288). It beat Logistic Regression in 99.8 percent of re-draws, but since Logistic Regression was itself below random on validation, this was not a real win; against random checking, it won only 43 percent of the time. Across six random seeds its score ranged from 0.231 to 0.293, with only two above random. On the test period it equalled random checking (0.232). Under the rule set in advance, Logistic Regression stayed as the application model: it is simpler, faster and fully explainable.

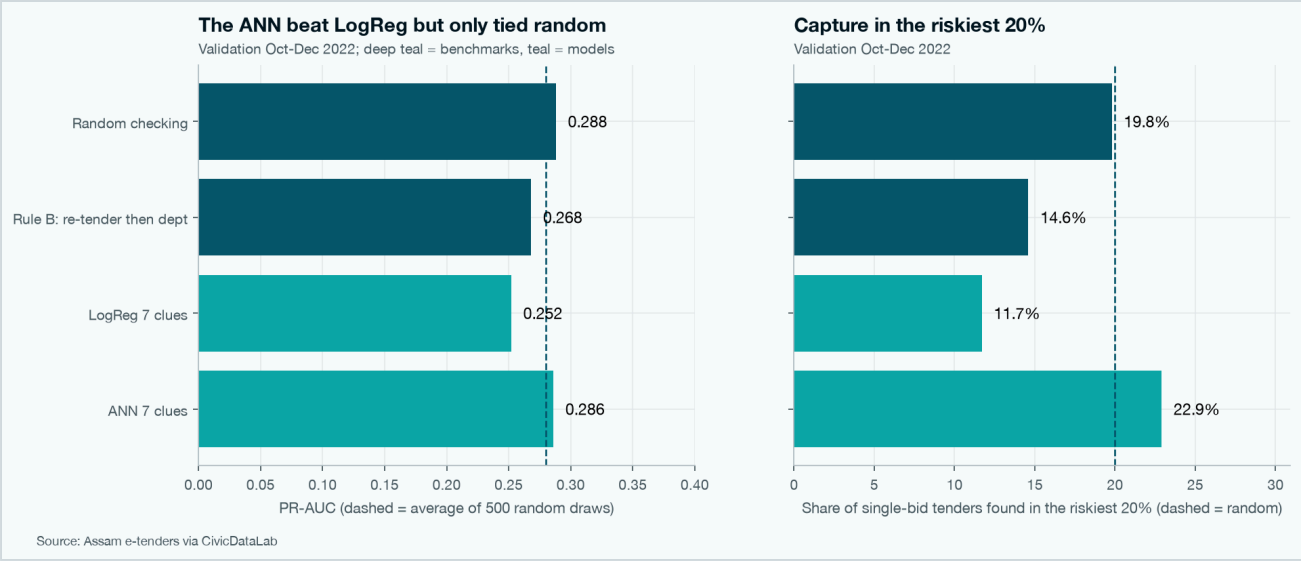


Figure 24 — Validation results: the neural network only tied random checking

11.7 Unusual tenders overlap with high-risk ones, and reveal data errors

The Isolation Forest flagged 74 test tenders as unusual. Fifty-four of them (73 percent) were also in the High risk band, so the two signals agree more than expected. The flag is therefore used to add a reason ("which clue is odd") rather than as a second risk score. Unusual tenders were mostly large works with fewer bidders (median value Rs 697 lakh against Rs 94 lakh). The checks also found likely data-entry errors, such as a solar power plant (tender 2023_APGCL_29273_1) recorded at a value of Rs 3,000.

Combining both signals in the check-first rule picked 246 test tenders as Priority 1: about one in seven tenders, with a single-bid rate of 30.9 percent against 23.8 percent overall, and 16.9 percent of the test period's money.

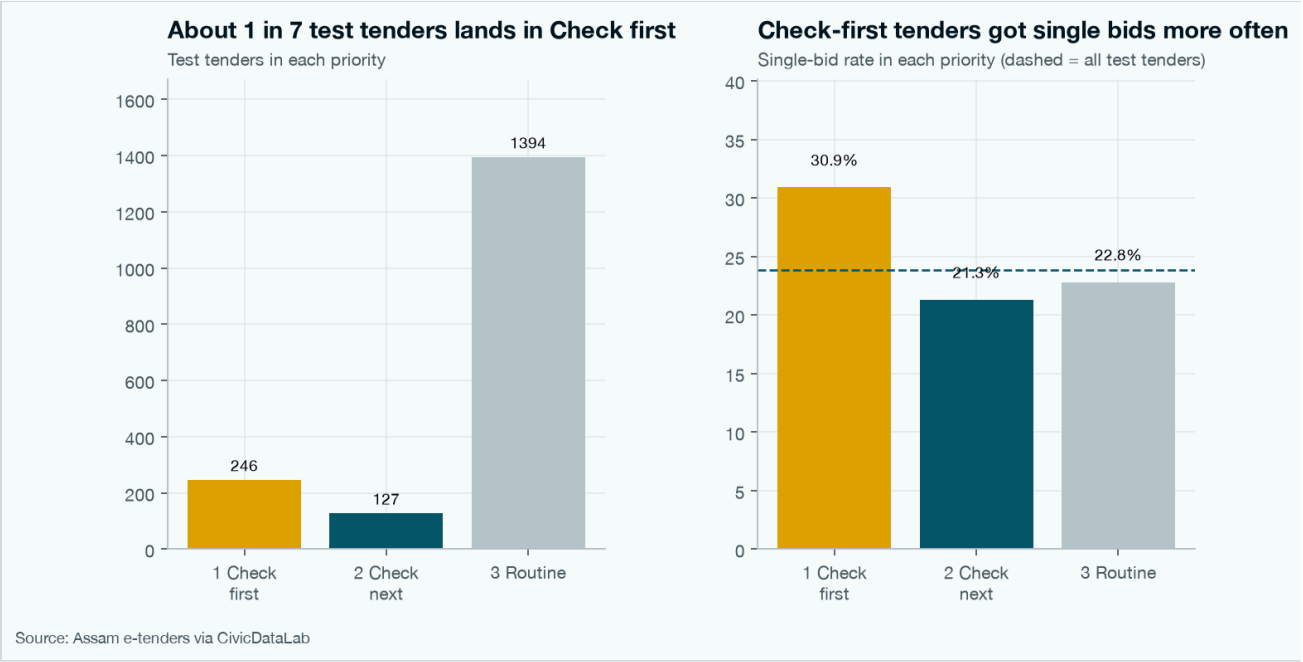


Figure 25 — The check-first rule: Priority 1 tenders have the highest single-bid rate

11.8 Re-posting a tender does not bring more bidders, and some departments move against the trend

Among 507 tenders whose title was posted on two or more different days, the first attempt was single-bid 19.2 percent of the time, while second attempts were single-bid 33.3 percent of the time and third or later attempts 30.0 percent. Simply re-posting the same tender did not attract more competition. At department level, Bodoland Territorial Council went against the overall improvement, rising from 24.7 to 39.9 percent single-bid after September 2022, and Guwahati Municipal Corporation rose from 1.9 to 31.9 percent. Because the model learned these departments' older, lower rates, they explain where it missed on test. This is exactly why the application includes a model health page and monthly retraining.

12. Challenges Faced

A number of genuine obstacles came up during the project, and each one is recorded here rather than left out, since working through them is as much a part of the project as the final model.

- **Unusual file structure.** Each tender CSV mixed one row of tender details with extra rows of lists, and awarded tenders had two bidder name columns with different meanings. The bid count could only be built correctly after studying many files by hand.
- **Messy values.** Negative bidding windows, values of Rs 0, values above Rs 1,000 crore, award ratios far outside a sensible range, and 454 spellings of department names all had to be handled with clear, documented rules.
- **A gap in the data.** The portal data had a gap between March 2021 and February 2022, which could easily have been mistaken for a real fall in tenders. It was marked instead.
- **Few labelled tenders.** Only 6,790 of 39,411 tenders had a bidder list, and most of them came from 2022, so the model had a short history to learn from.
- **Drift.** Every model, including XGBoost and the neural network, failed to beat random checking on validation. Instead of tuning until the numbers looked good, the cause was investigated and documented as a finding.
- **Label timing leak.** A monthly retraining plan first looked like the best result in the project. Closer checking showed it was learning from tenders whose bids were not yet open. Fixing this removed the gain, and the "opened only" rule was applied from then on.
- **Text model download.** The MiniLM sentence model could not be downloaded in the cloud test environment, so the embeddings were created on the laptop, saved, and then reused.
- **Library version clashes.** SHAP, Numba and TensorFlow needed NumPy 1.26, while some newer libraries tried to install NumPy 2. Versions were pinned (for example NumPy 1.26.4, SHAP 0.46.0 and pyarrow below 20) so that every notebook and the application run reliably.
- **XGBoost on Mac.** XGBoost needed an extra system library (libomp), installed through Homebrew.
- **Neural network early stopping.** Without balanced weights, the early stopping signal stopped training after one or two rounds. Balanced sample weights fixed this. SHAP also clashed with Keras 3, so permutation importance was used to explain the neural network.
- **Messy location field.** The location column had 1,584 different texts and only 64.5 percent matched a district name, so a district-level analysis was checked and then dropped rather than reported on unreliable data.

13. Real-Life Applications

The findings and the TenderLens application have direct, practical uses beyond the scope of a classroom exercise:

- **Audit and vigilance teams** could use the monthly check-first list to decide which newly published tenders to review first, instead of choosing at random or by habit.
- **Department heads and procurement officers** could use the department report card to see how their single-bid rate and money at risk compare with other departments, and act before bids close, for example by extending the bidding window or advertising more widely.
- **State finance departments and audit offices** could use the money-at-risk and competition-price findings to target departments where single bidding concentrates the most public value.
- **Data quality teams** could use the unusual-tender flags and data quality alerts to correct entry errors, such as a large project recorded at a few thousand rupees, at the source.
- **Researchers, journalists and civil society groups** could use the open pipeline as a starting point for studying competition in public procurement in other states, without repeating the cleaning and validation work.

14. Limitations

Every dataset and every model has boundaries, and this project's are stated here directly rather than left implicit.

- Only 6,790 tenders have a verified bidder list, and most are from 2022. The model therefore learns from a short period, and patterns before 2022 are barely represented.
- Tender patterns drifted strongly after September 2022. Any model trained on the past will weaken as conditions change, which is why monthly retraining and a model health page are part of the design.
- The improvement over random checking is real but small (101 against 79 tenders in the riskiest 20 percent), and a simple rule (Rule B) did slightly better on test. The model should be read as one useful input, not a final answer.
- Publish-day clues are weak by nature. The strongest drivers of single bidding, such as local market conditions or how widely a tender was advertised, are not recorded in the data.
- Risk scores rank tenders but are not true probabilities, so the application shows bands rather than percentages.
- The cost-of-one-bidder estimate (Rs 120 to 680 crore) and the price comparisons are signs, not proof, because single-bid tenders may differ from multi-bid tenders in ways the data does not capture.
- The location field was too inconsistent to use, so the project cannot compare districts.
- A high risk score never means wrongdoing. TenderLens is screening support only.

15. Recommendations

The project moves deliberately from analysis to action. Four priorities follow directly from the data rather than from general assumptions about procurement.

15.1 Review tenders from a monthly check-first list

Instead of random checks, reviewers should start each month with the Priority 1 tenders: those in the High risk band that are also unusual or worth at least Rs 1 crore. On the test period this was about one tender in seven, with a single-bid rate of 30.9 percent against 23.8 percent overall. Because tenders are ranked within their own month, the list stays fair even when one month is much busier than another.

15.2 Aim for four or more bidders on large works

Real price discounts appeared only when four or more firms bid. For high-value tenders flagged as High risk, departments should act before the bidding closes, for example by allowing a longer bidding window, advertising more widely, or splitting very large works into sensible lots, with the goal of reaching four or more bidders rather than just two.

15.3 Watch the departments that move against the trend

Most departments improved after September 2022, but Bodoland Territorial Council and Guwahati Municipal Corporation got worse, and three departments hold three quarters of single-bid money. A monthly department report card, as built into the application, should be reviewed with these departments so that changes are noticed early rather than after the year closes.

15.4 Retrain monthly and fix data at the source

The model should be retrained every month using only tenders whose bids have already been opened, and its health should be tracked against random checking. At the same time, the portal should require a tender value and a standard district field, and should block obvious entry errors such as a large project valued at a few thousand rupees. Better data would make every future model stronger.

> *These four priorities target where competition fails and why, not just the symptoms visible after the bids open.*

16. Conclusion

This project set out to answer a specific question: can Assam government tenders that are likely to receive only one bid be identified on the day they are published, and can the reasons be explained clearly enough for a reviewer to act on them? The answer is a qualified yes. On a fair, one-time test on future tenders, the final model found 101 single-bid tenders in its riskiest 20 percent, against 79 by random checking, and this improvement held in 95.8 percent of 500 re-draws. It is a small gain, but it is a real one, and it was reached without any leakage or test-set tuning.

The more important lesson is about honesty in machine learning on real government data. Tender patterns changed sharply after September 2022, and several results that first looked strong, including a monthly retraining plan and the neural network, did not survive a fair test. Each of these was reported rather than hidden, and the simpler, explainable Logistic Regression was kept because the more complex options did not earn their place. Beyond the model, the data tells a clear story: single-bid tenders carry 36.5 percent of the money, real discounts only appear with four or more bidders, and re-posting a tender does not by itself bring more competition.

The project also demonstrates a complete, real-world data science workflow: sourcing and cleaning a large and messy government dataset, exploring it with SQL and Python, engineering leakage-safe features, comparing classical, text and deep learning models against honest benchmarks, explaining predictions with SHAP, detecting anomalies, and delivering everything through an interactive application with a responsibly designed GenAI feature. TenderLens does not replace human judgement; it helps reviewers decide where to look first.

17. Future Scope

A number of enhancements were identified during the project but placed outside its current scope, either because of time or because they depend on data not available here. They are recorded as a roadmap.

- Deploying the application on Streamlit Community Cloud with a live LLM key, so that reviewers can use it from any browser.
- Extending the analysis to other states whose tender data is published in a similar format, to test whether the same patterns and clues hold elsewhere.
- Adding a feedback loop, where reviewers mark checked tenders and their findings are used to improve future rankings.
- Retraining automatically every month and alerting the reviewer when drift crosses a set level.
- Using the full tender description once the portal stops cutting text at 250 characters, which may make the text methods more useful.
- Cleaning the location field into standard district names, so that district-level patterns can be studied.
- Studying firms' bidding networks over time (with names kept anonymous), to understand why some markets attract only one bidder.

18. References

- CivicDataLab. Assam Tenders Data, 2016–2023. Public GitHub repository, github.com/CivicDataLab/assam-tenders-data. Original source: Government of Assam e-tenders portal, assamtenders.gov.in.
- OECD (2016). Preventing Corruption in Public Procurement. OECD Publishing, Paris.
- Fazekas, M. and Kocsis, G. (2020). Uncovering High-Level Corruption: Cross-National Objective Corruption Risk Indicators Using Public Procurement Data. British Journal of Political Science, 50(1), 155–164.
- Lundberg, S. M. and Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. Advances in Neural Information Processing Systems 30 (SHAP).
- Liu, F. T., Ting, K. M. and Zhou, Z.-H. (2008). Isolation Forest. Proceedings of the IEEE International Conference on Data Mining.
- Reimers, N. and Gurevych, I. (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. Proceedings of EMNLP 2019 (all-MiniLM-L6-v2 model).
- Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.
- Chen, T. and Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. Proceedings of KDD 2016.
- TensorFlow and Keras documentation, tensorflow.org; Streamlit documentation, docs.streamlit.io.
- Project code and notebooks: github.com/sindhusharmamarupaka